

ARC and elabFTW Introduction IMET

June, 2026

Sabrina Zander (MibiNet)

Tentative agenda

Morning

Time	Topics
09:30 - 09:45	Welcome, Intro RDM and NFDI
09:45 - 10:15	Overview DataPLANT, and DataPLANT Tool chain
10:15 - 10:45	Hands-on Start Here Guide Part I
10:45 - 11:00	<i>Short break</i> 🍰
11:00 - 11:30	Metadata Intro
11:30 - 12:15	Hands-on Start Here Guide Part II

Afternoon

Time	Topics
12:15 - 13:15	<i>Lunch break</i> 🍕
13:15 - 14:15	Hands-on Start Here Guide Part III
14:15 - 14:30	DataHUB Intro
14:30 - 15:00	Hands-on Start Here Guide Part IV
15:00 - 15:30	Q & A ARC
15:30 - 15:45	<i>Short break</i> 🍰
15:45 - 16:15	elabFTW Intro
16:15 - 16:45	elab2ARC tool
16:45 - 17:30	Q & A and wrap-up

House-keeping

- Use the [pad](#) to raise questions and feedback
- Copy / paste links (hands-on)



Training Materials

Slides are shared via [DataPLANT Training Material](#)

Resources – join the open source movement



DataPLANT (nfdi4plants)

DataPLANT Website: <https://nfdi4plants.org/>

Knowledge Base: <https://nfdi4plants.org/nfdi4plants.knowledgebase/>

DataHUB: <https://git.nfdi4plants.org>

GitHub: <https://github.com/nfdi4plants>

HelpDesk: <https://helpdesk.nfdi4plants.org>

💡 You can help us by raising issues, bugs, ideas...



NEW! ARC website: <https://arc-rdm.org>

Continuous support

1. HelpDesk: <https://helpdesk.nfdi4plants.org>
2. Matrix workspace for ad hoc support: <https://matrix.to/#/%23arc-usersupport:matrix.org>
3. User Support Mailing List: arc-user-support-join@lists.nfdi.de
4. User Support Meeting (2nd Friday of the month | 1-2pm):
<https://nfdi4plants.github.io/events/arc-user-support/>

Introduce yourself

- Lab
- MibiNet / CEPLAS / TRR / ?
- Used code / programming language before
- Experience with Git / GitLab / GitHub?
- Have an ORCID
- My motivation / expectation
- My favorite lab assay

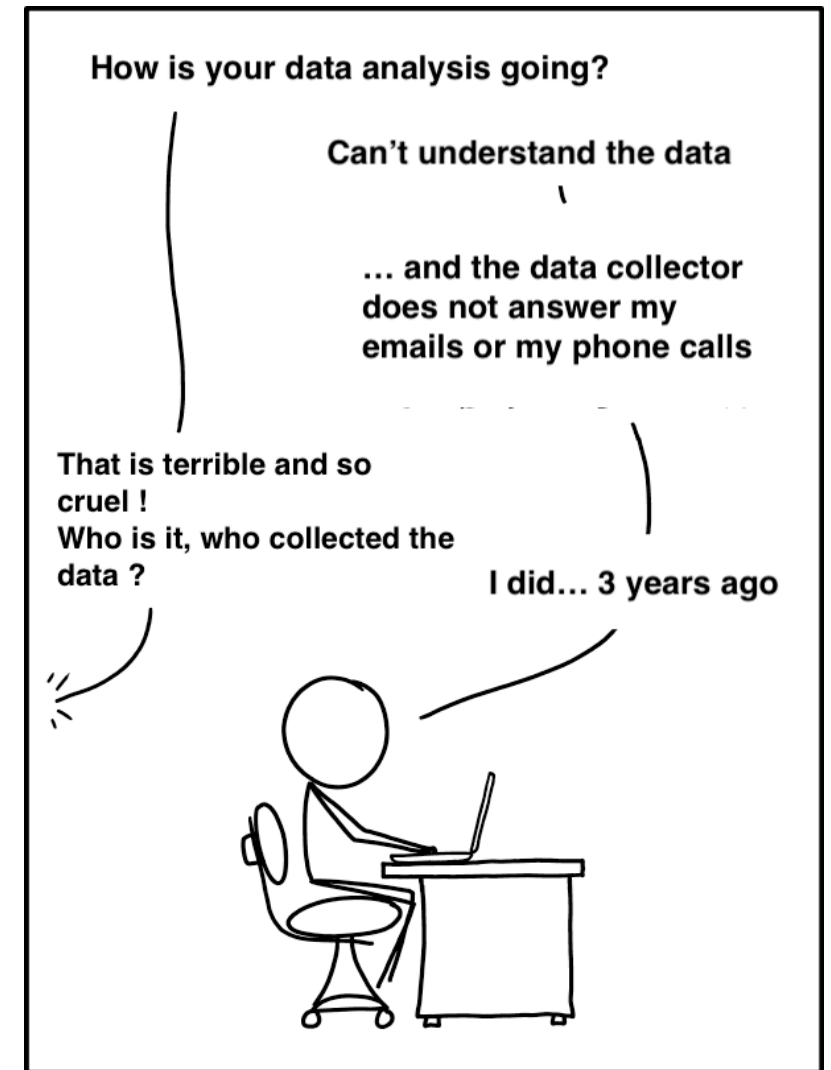
Let's draw a typical lab workflow

BYOD – Goals

- First few steps into ARC ecosystem
- Move demo datasets into ARCs
- Share them via the DataHUB
- Annotate with metadata
- Introduction into elabFTW
- elab2ARC tool

Why research data management (RDM)?

- Increase transparency
- Make data accessible
- Save time (writing, reusing)
- Reduce the risk of data loss
- Optimize the costs
- Facilitate future reuse and sharing
- Improve citations



**Your first collaborators
are your future selves,
be nice to them !**

your future self, by Julien Colomb, CC-BY-NC, derived from .NORM Normal File Format, CC-BY-NC, by Randall Munroe

FAIR data stewardship

- Findable
- Accessible
- Interoperable
- Reusable

<https://doi.org/10.1038/sdata.2016.18>

[nature](#) > [scientific data](#) > [comment](#) > [article](#)

[Open Access](#) | [Published: 15 March 2016](#)

The FAIR Guiding Principles for scientific data management and stewardship

[Mark D. Wilkinson](#), [Michel Dumontier](#), [IJsbrand Jan Aalbersberg](#), [Gabrielle Appleton](#), [Myles Axton](#), [Arie Baak](#), [Niklas Blomberg](#), [Jan-Willem Boiten](#), [Luiz Bonino da Silva Santos](#), [Philip E. Bourne](#), [Jildau Bouwman](#), [Anthony J. Brookes](#), [Tim Clark](#), [Mercè Crosas](#), [Ingrid Dillo](#), [Olivier Dumon](#), [Scott Edmunds](#), [Chris T. Evelo](#), [Richard Finkers](#), [Alejandra Gonzalez-Beltran](#), [Alasdair J.G. Gray](#), [Paul Groth](#), [Carole Goble](#), [Jeffrey S. Grethe](#), [Jaap Heringa](#), [Peter A.C. 't Hoen](#), [Rob Hooft](#), [Tobias Kuhn](#), [Ruben Kok](#), [Joost Kok](#), [Scott J. Lusher](#), [Maryann E. Martone](#), [Albert Mons](#), [Abel L. Packer](#), [Bengt Persson](#), [Philippe Rocca-Serra](#), [Marco Roos](#), [Rene van Schaik](#), [Susanna-Assunta Sansone](#), [Erik Schultes](#), [Thierry Sengstag](#), [Ted Slater](#), [George Strawn](#), [Morris A. Swertz](#), [Mark Thompson](#), [Johan van der Lei](#), [Erik van Mulligen](#), [Jan Velterop](#), [Andra Waagmeester](#), [Peter Wittenburg](#), [Katherine Wolstencroft](#), [Jun Zhao](#) & [Barend Mons](#) 

[— Show fewer authors](#)

[Scientific Data](#) **3**, Article number: 160018 (2016) | [Cite this article](#)

The FAIR principles



Findable



Accessible

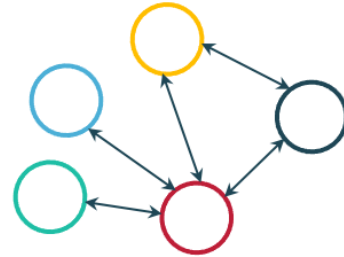


Interoperable



Reusable

Easier collaboration & sharing



Increased findability and visibility



Reproducibility



**Added-value to the
research community**

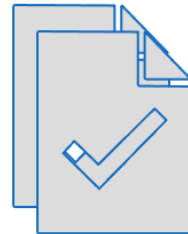
nfdi



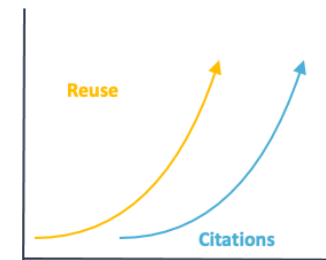
EMBL-EBI



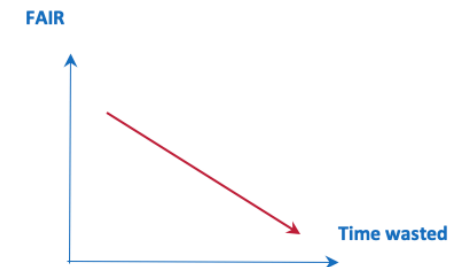
**Compliance with
funding policies**



Receive due credit



Saves time & workload



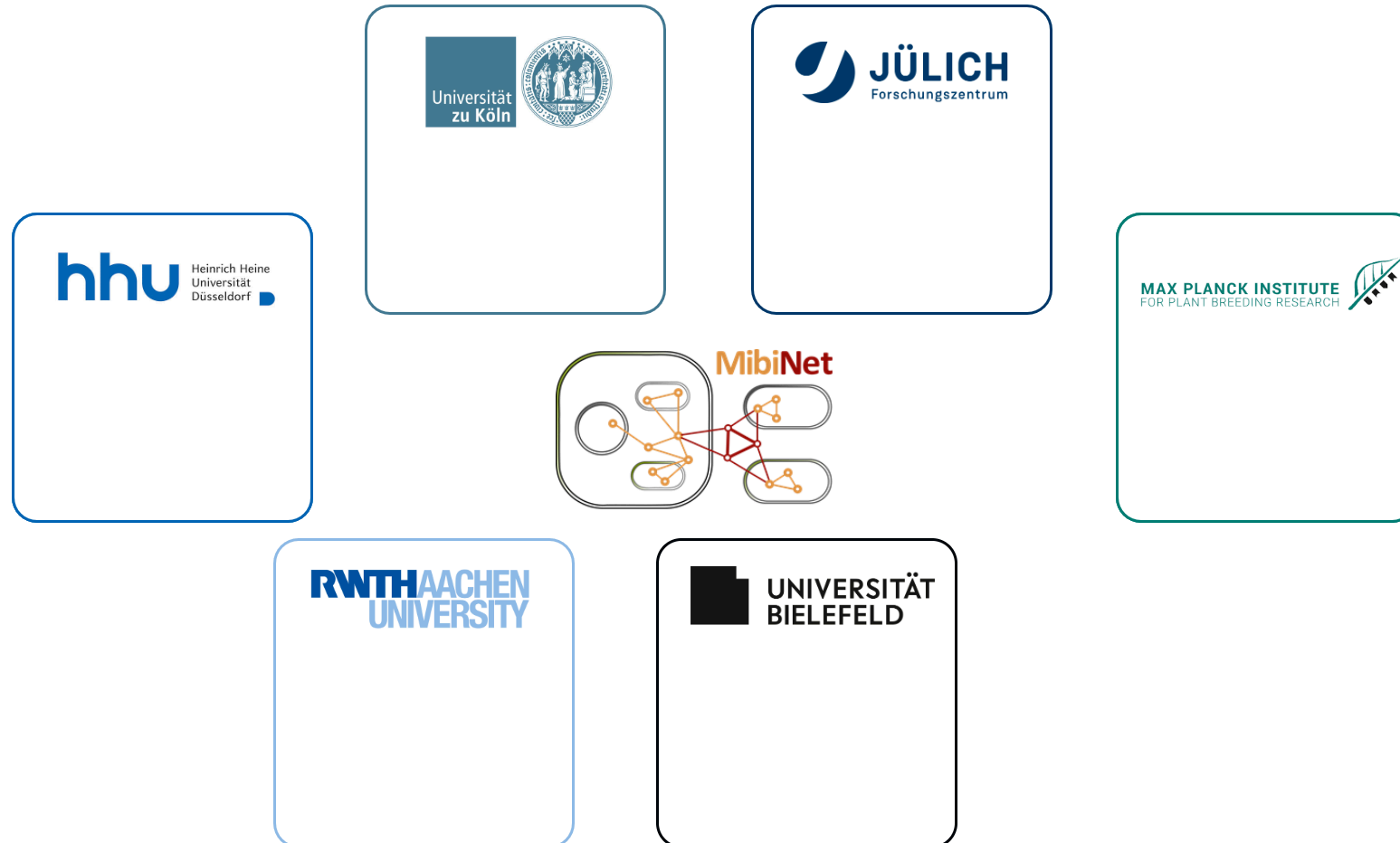
Is your data FAIR?

Findable | Accessible | Interoperable | Reusable

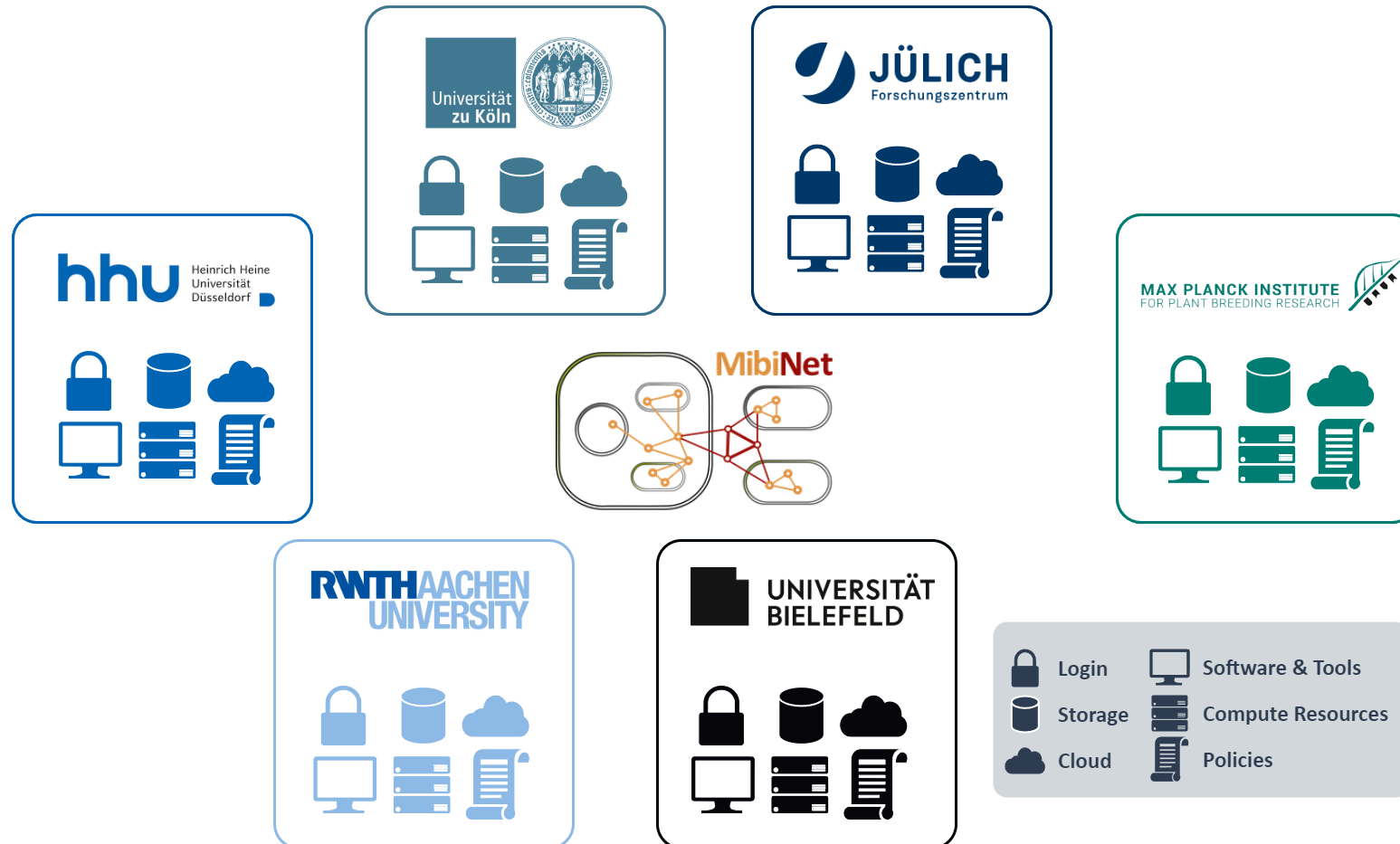
- Where do you store your data?
- How do you annotate your data?
- How do you share your data?
- What tools do you use to analyse your data?
- How do you reuse other people's data?



MibiNet – One SFB, six locations

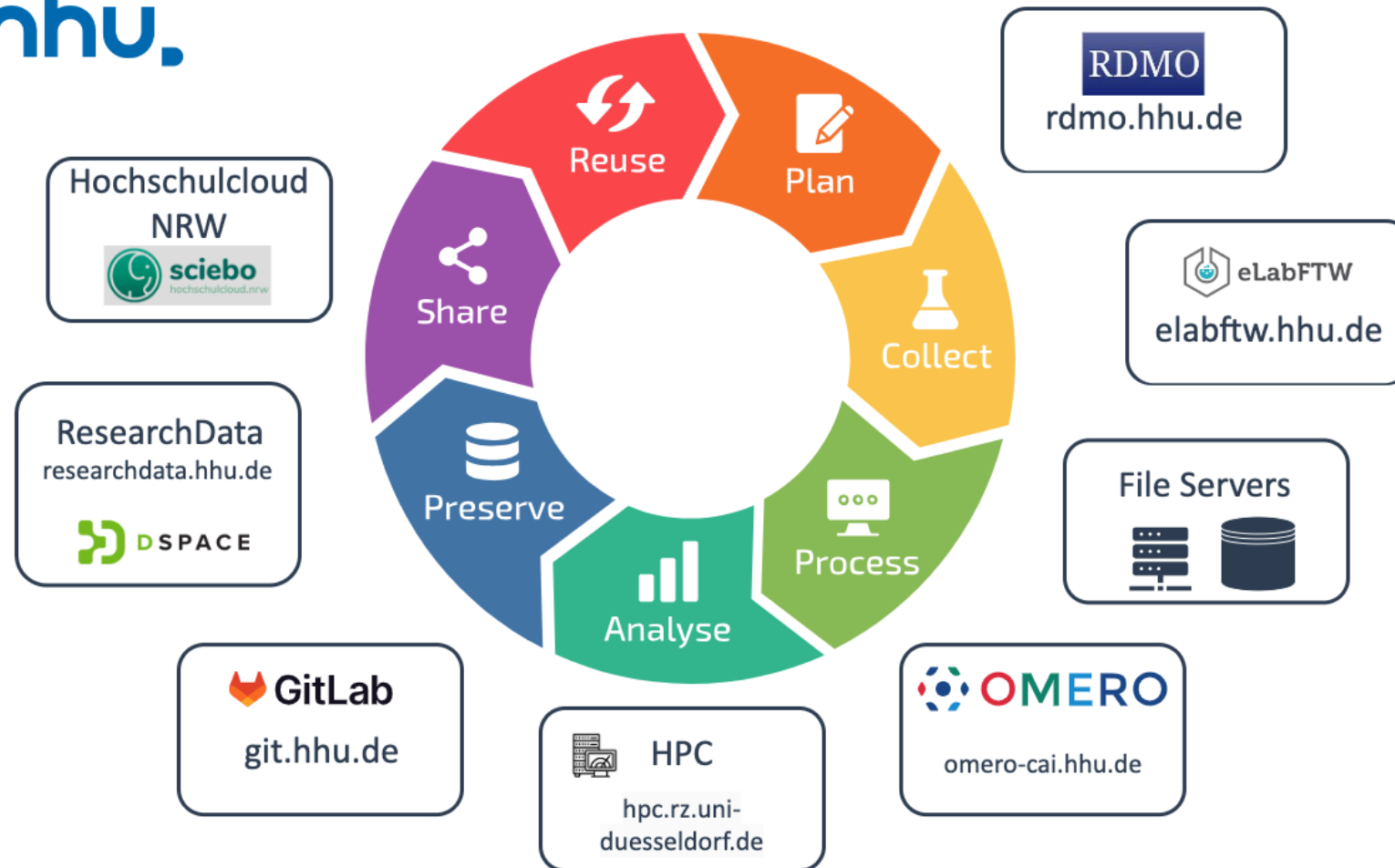


Data silos impede collaboration



Missing interfaces impede collaboration

hhu.



NFDI – Nationale Forschungsdaten Infrastruktur

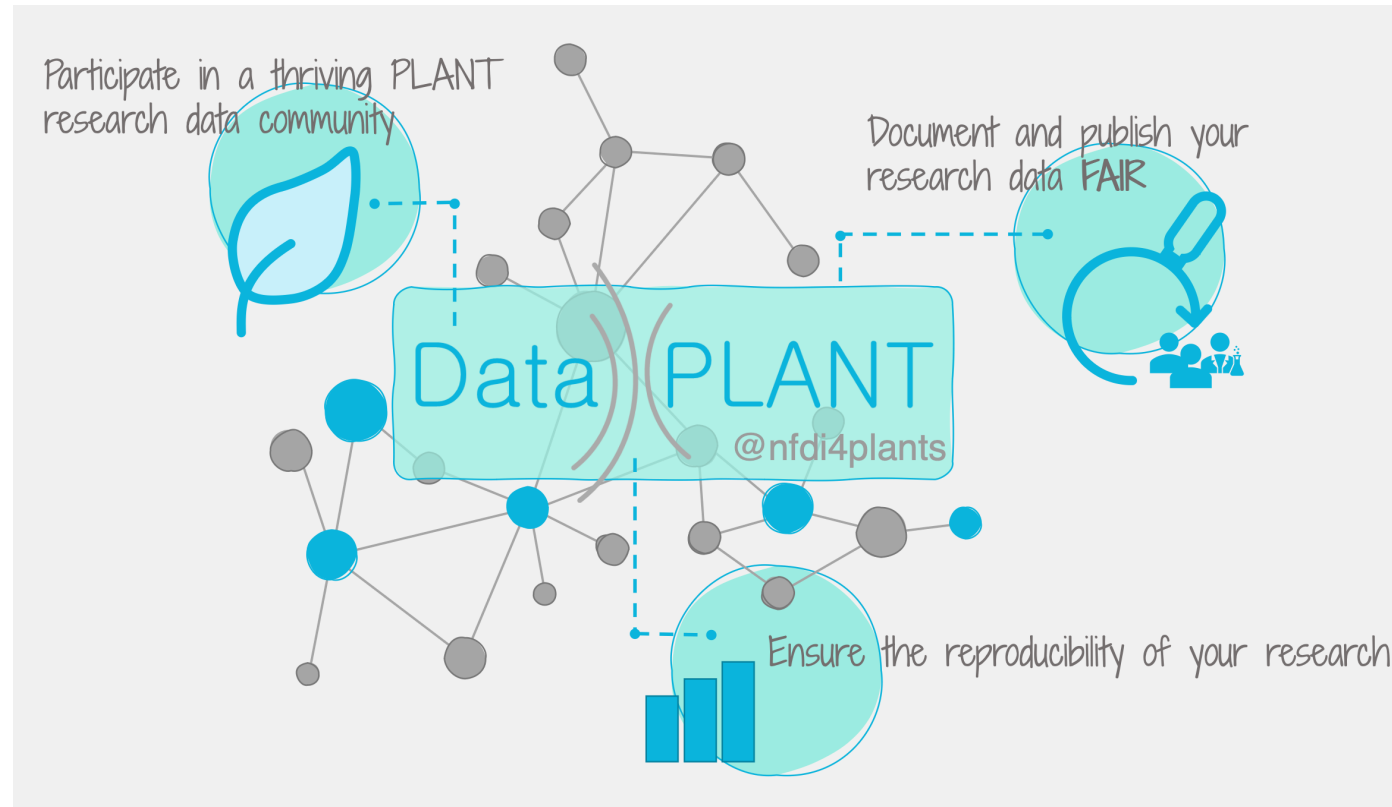


The aim of the national research data infrastructure (NFDI) is to systematically manage scientific and research data, provide long-term data storage, backup and accessibility, and network the data both nationally and internationally.

MibiNet connection to NFDIs

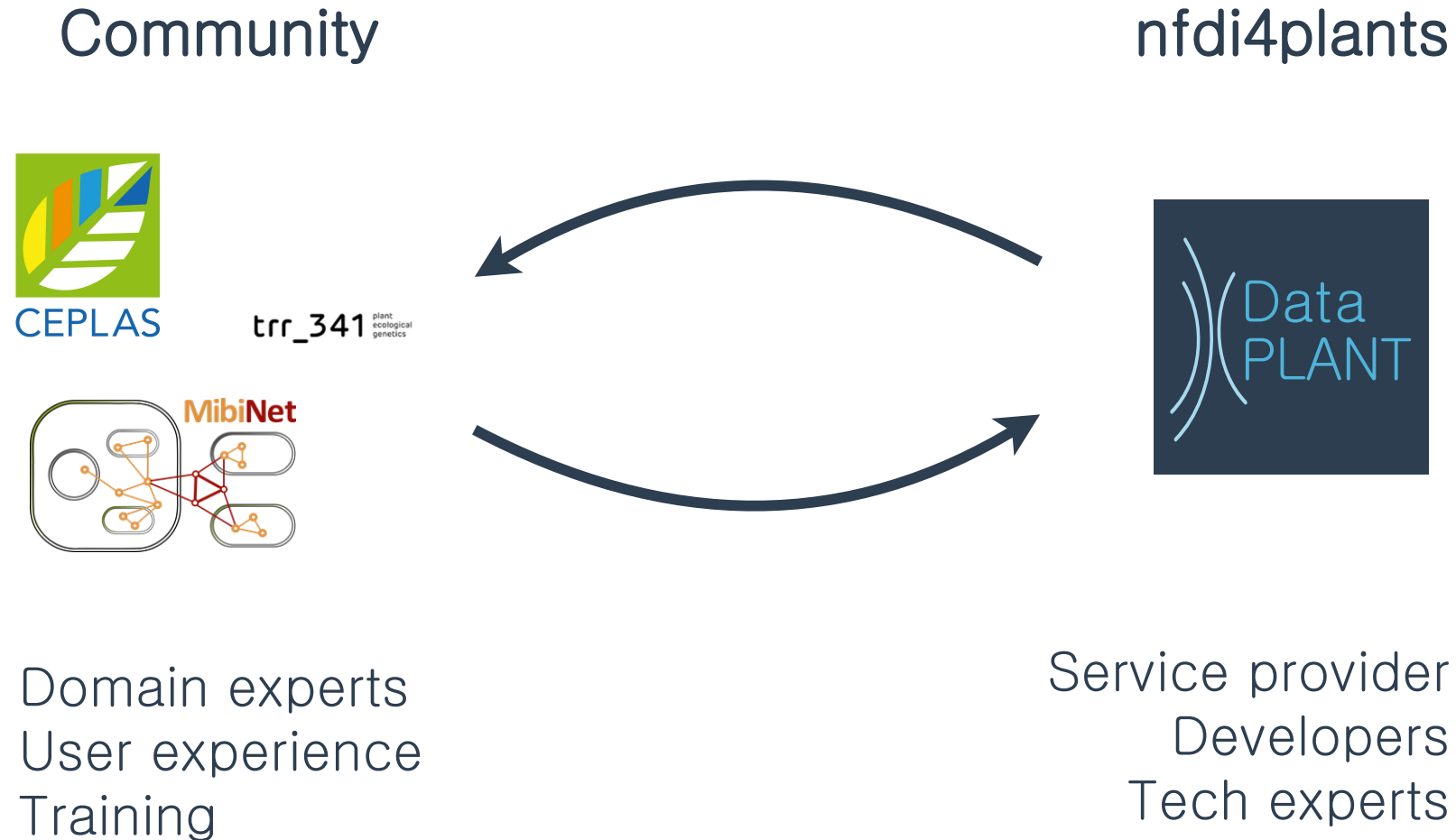


DataPLANT – *NFDI4plants*

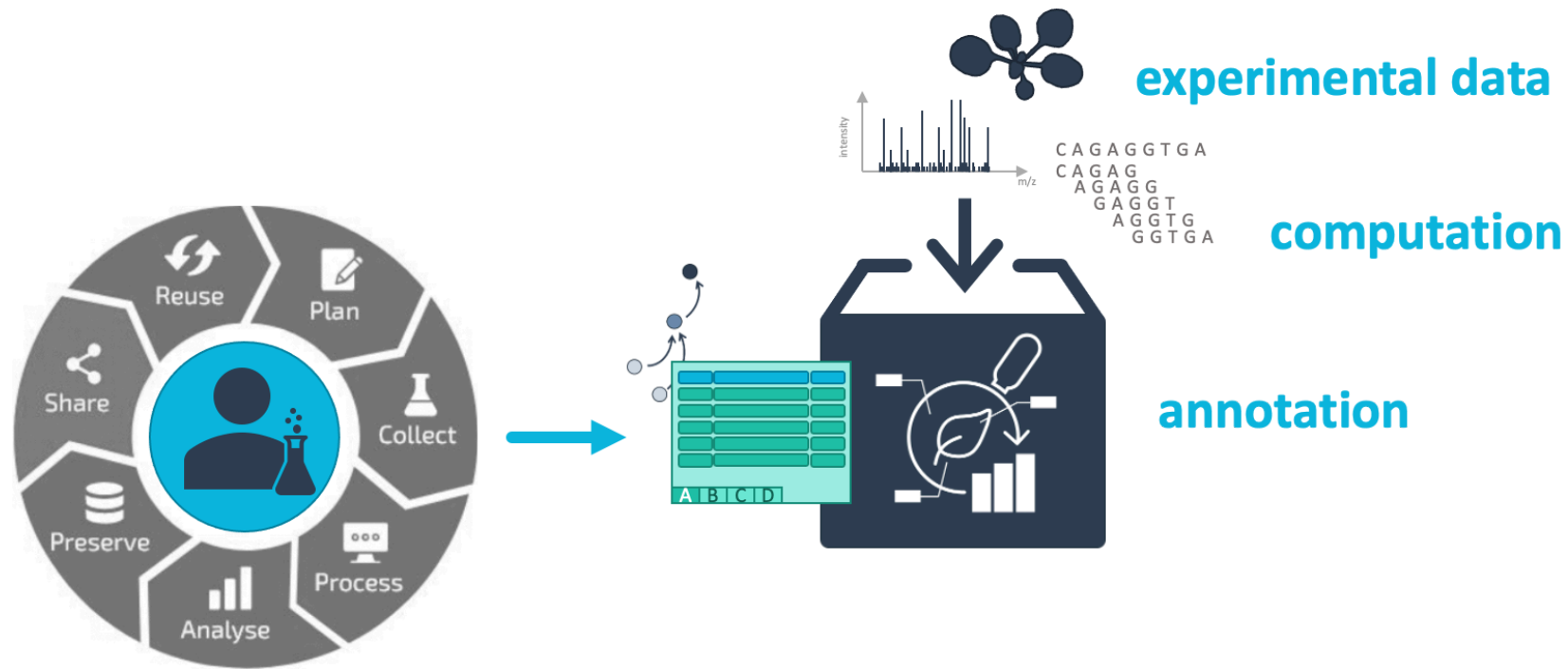


- <https://nfdi4plants.org>
- <https://arc-rdm.org>

Data Stewardship between DataPLANT and the community

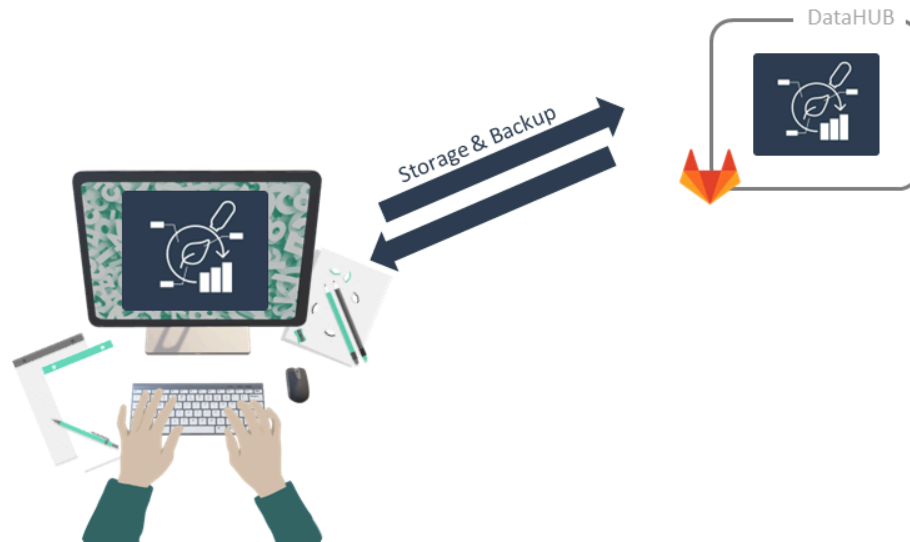


Annotated Research Context (ARC)

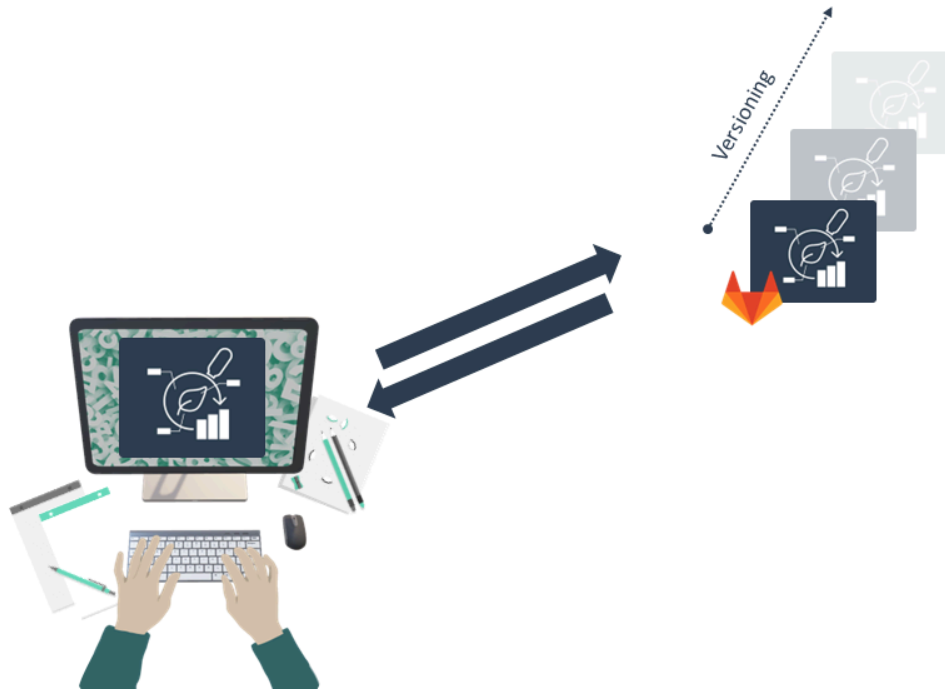


Your entire investigation in a single unified bag

You can store your ARC in the DataHUB



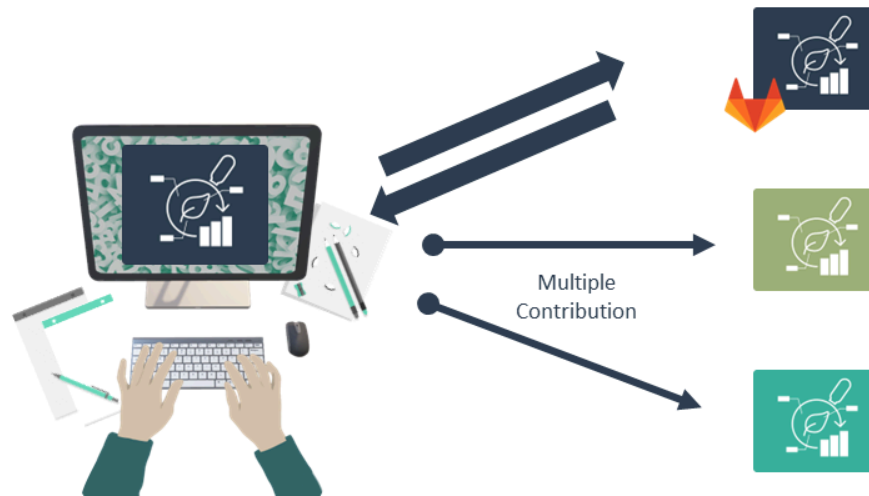
ARCs are versioned



You can invite collaborators



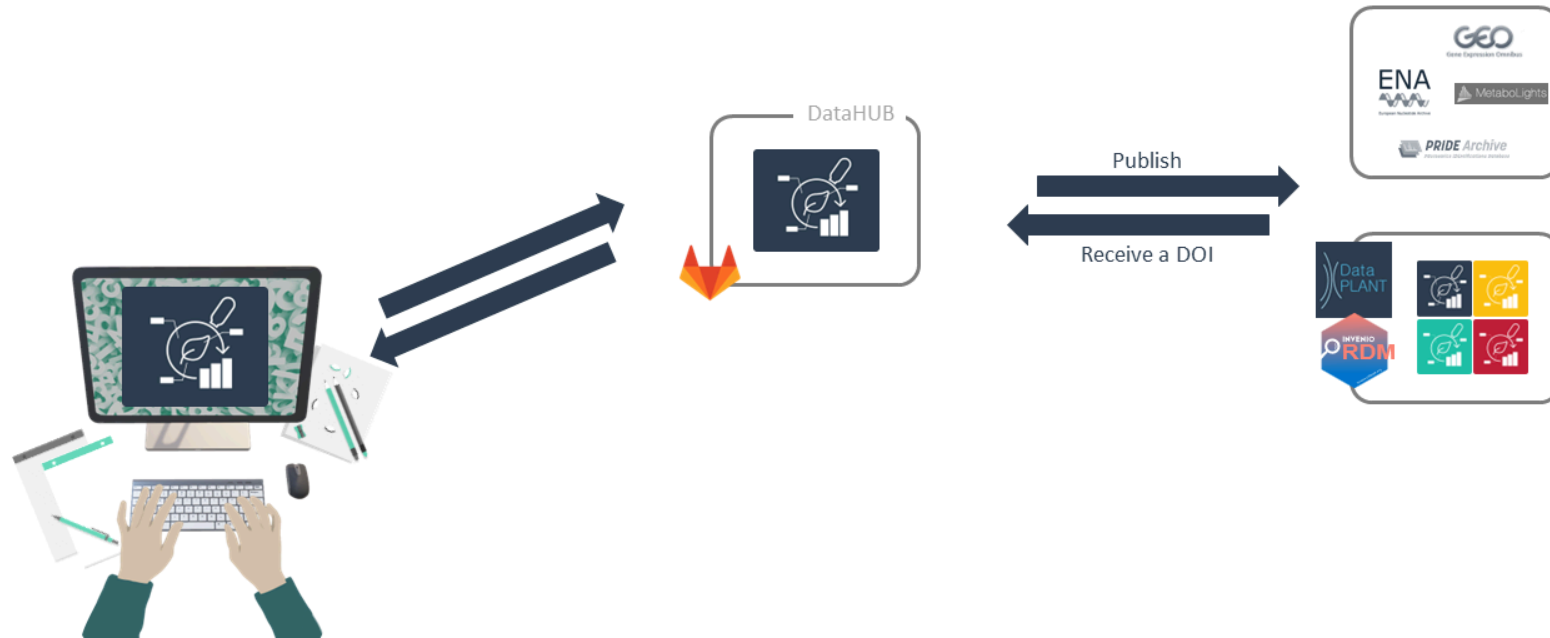
Collaborate and contribute



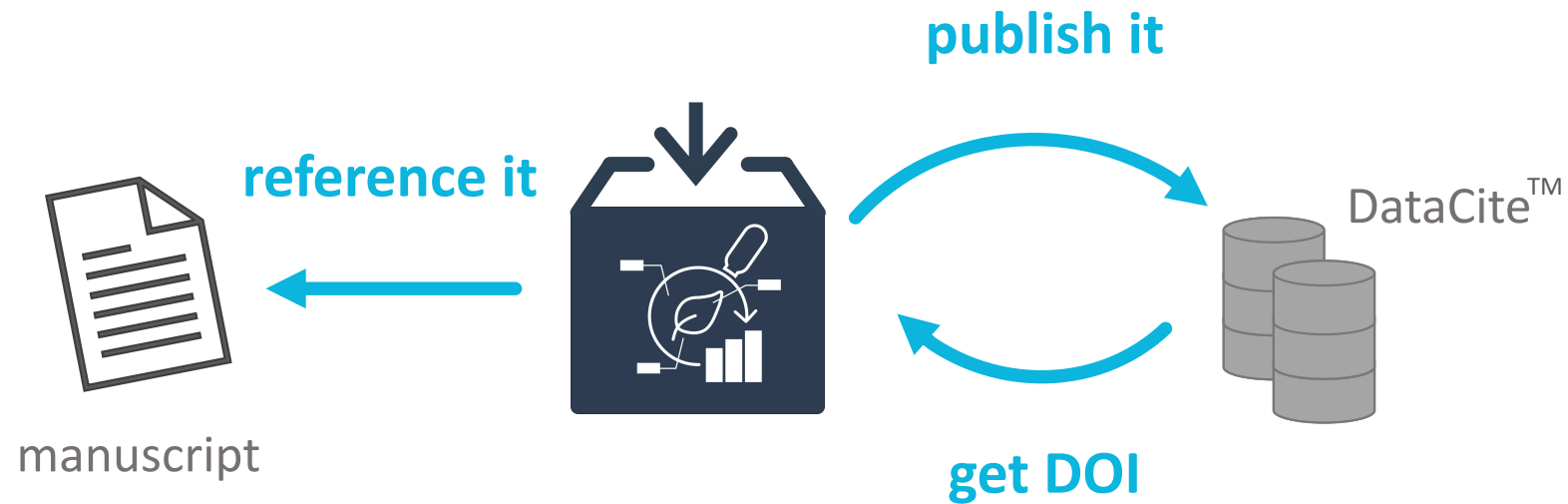
Reuse data in ARCs



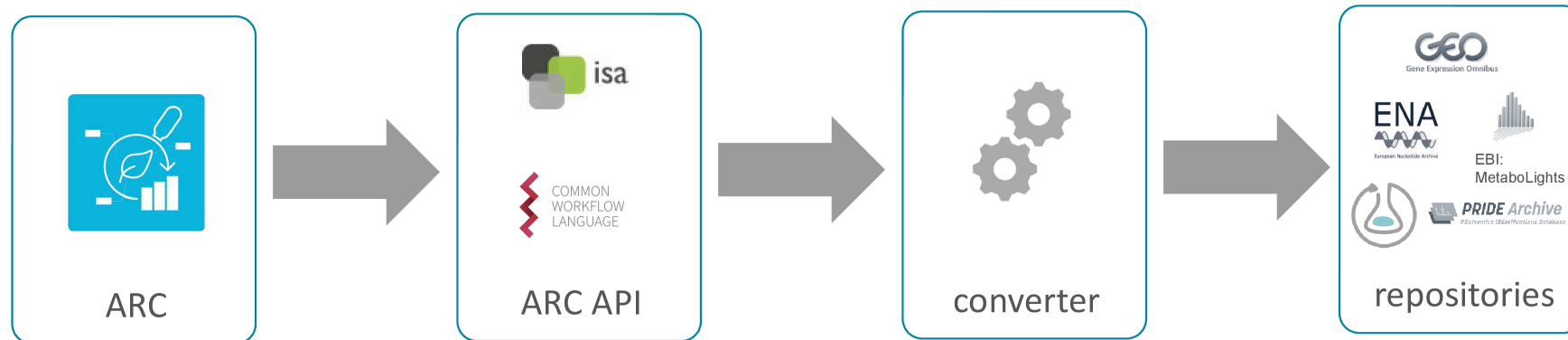
Publish your ARC



Publish your ARC, get a DOI



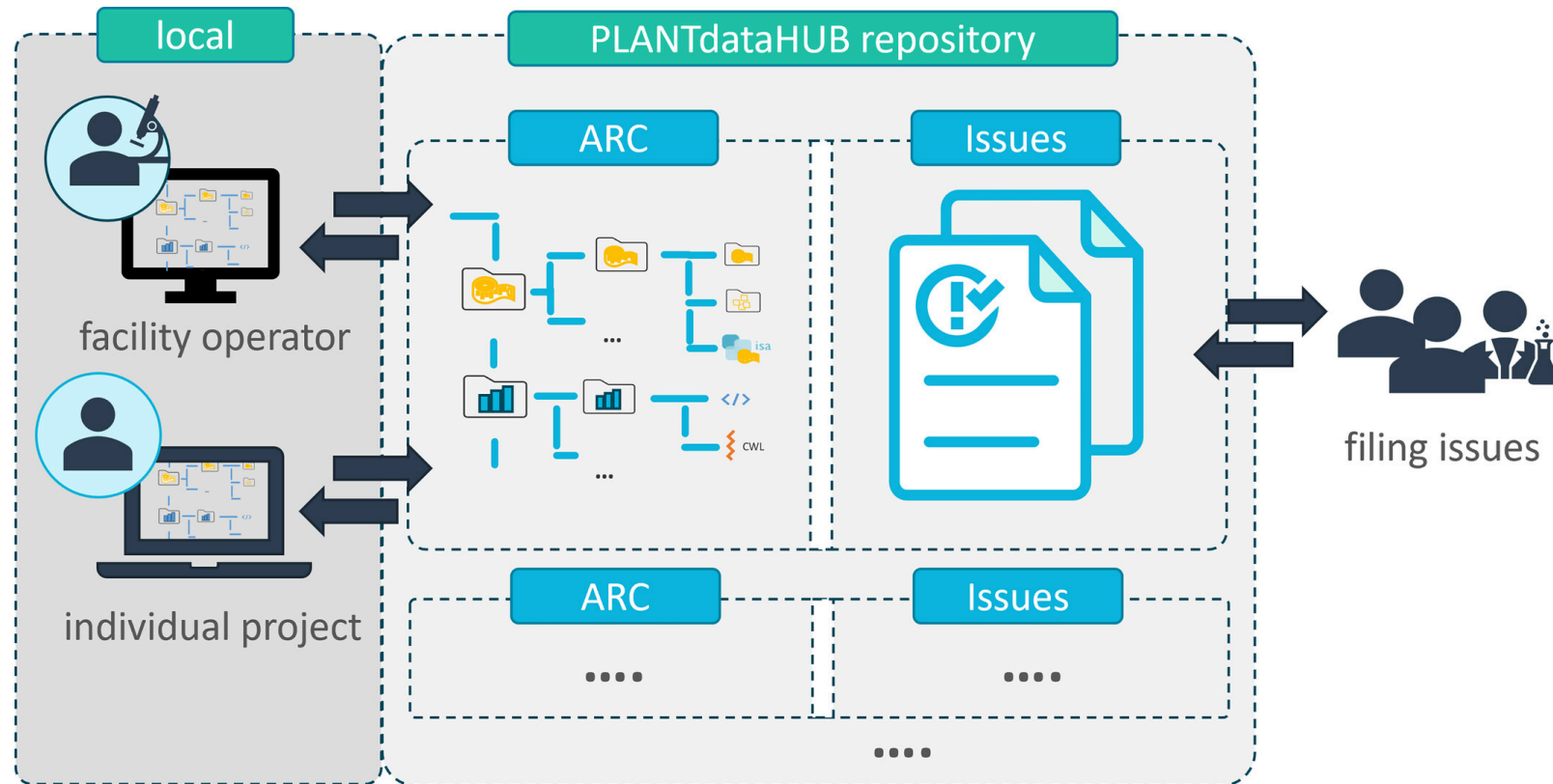
From ARC to repositories



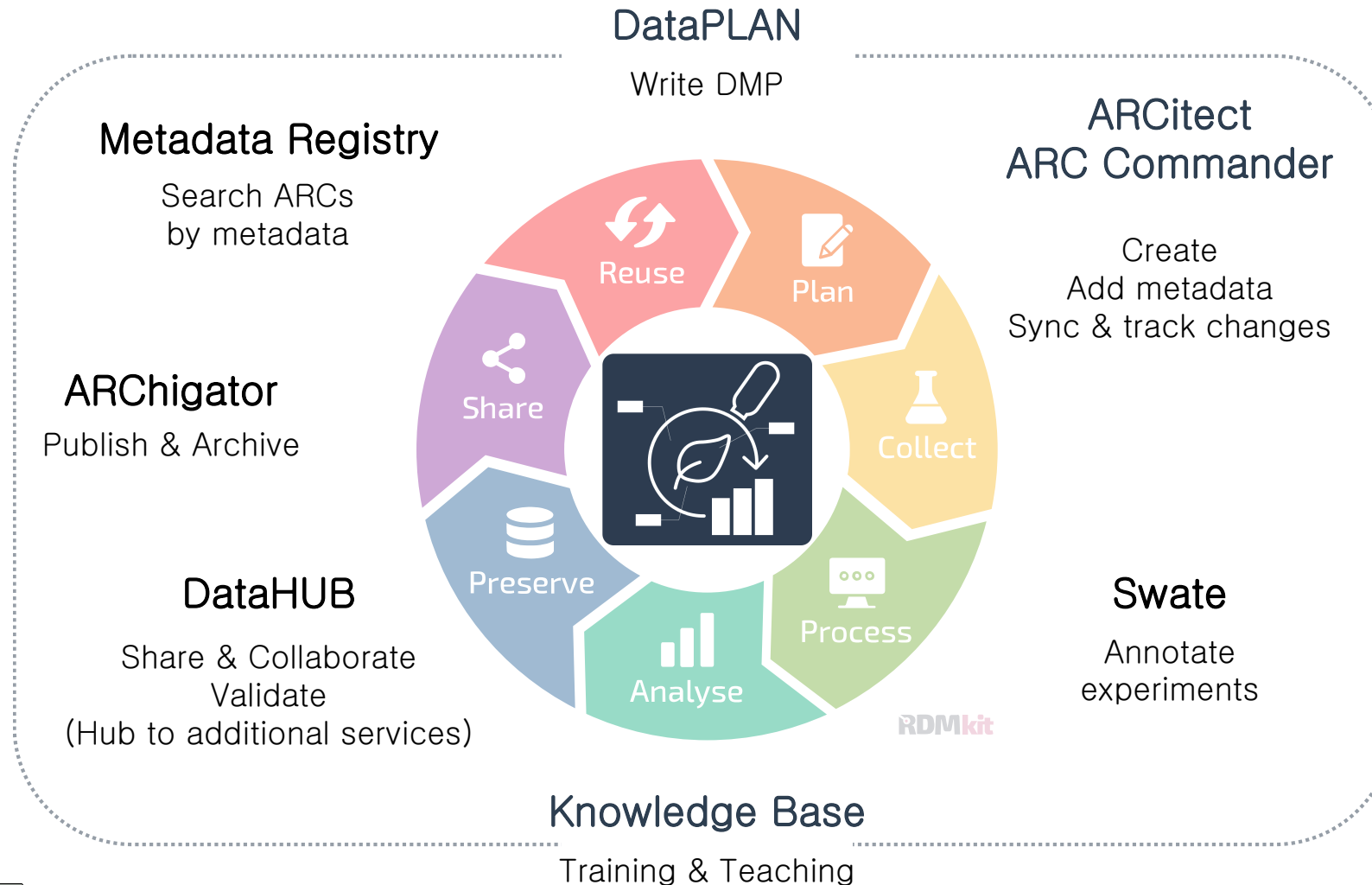
Moving from paper to data publications



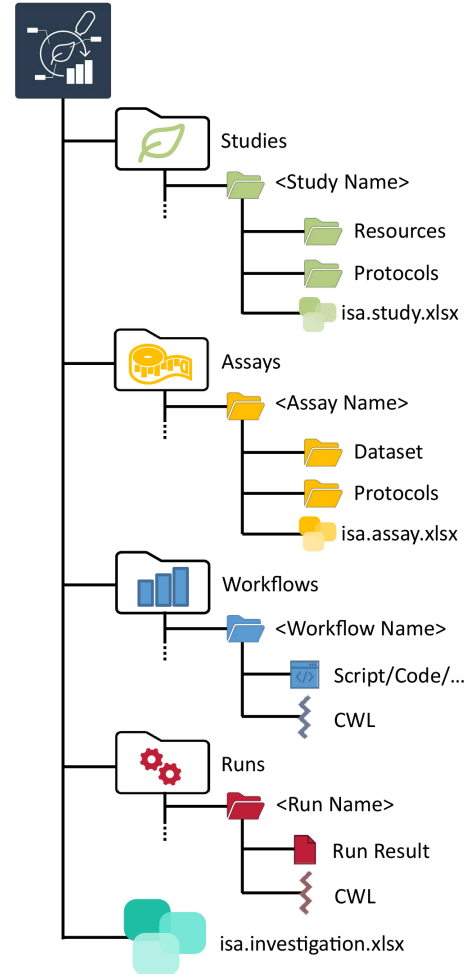
Project management



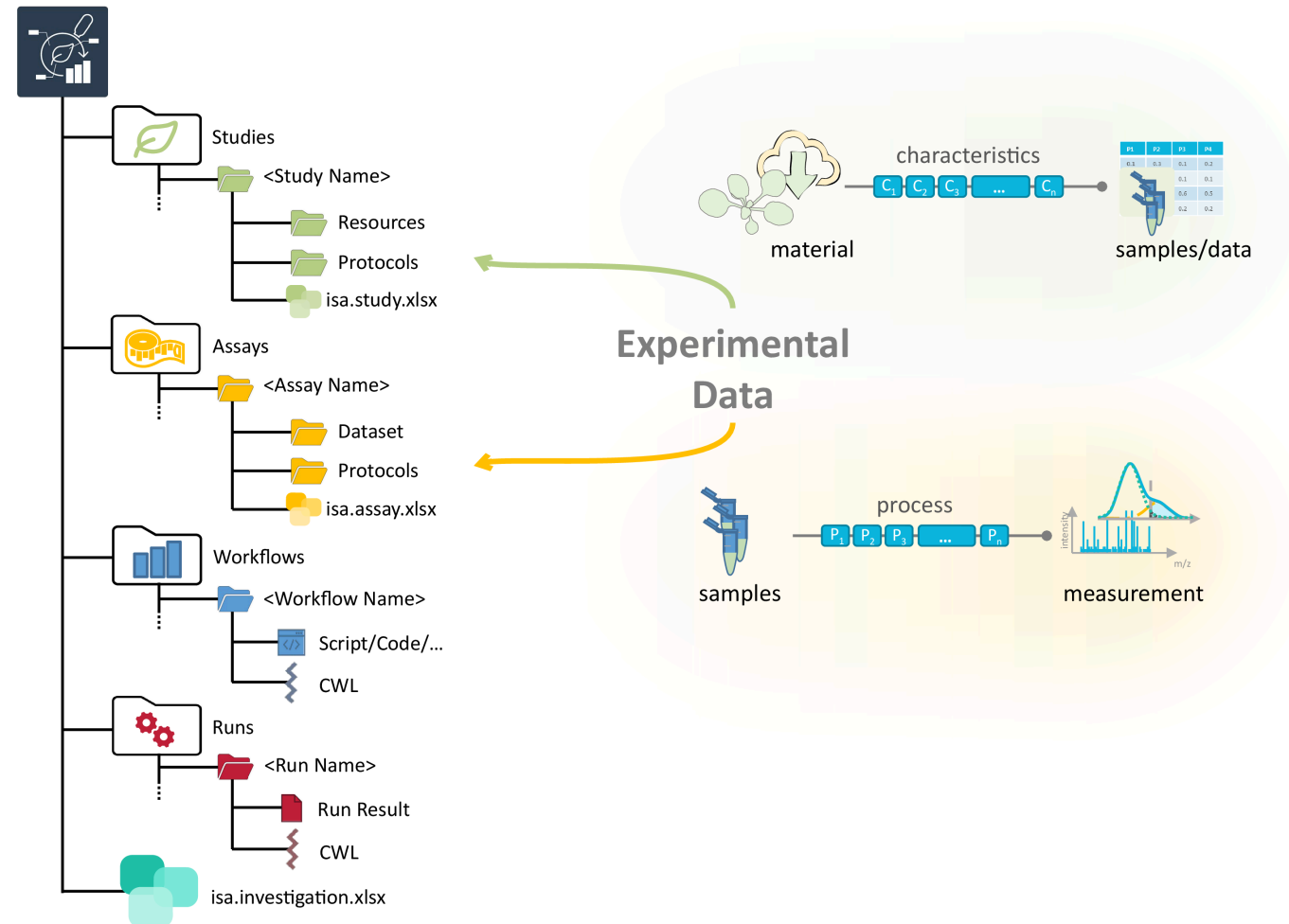
The ARC ecosystem



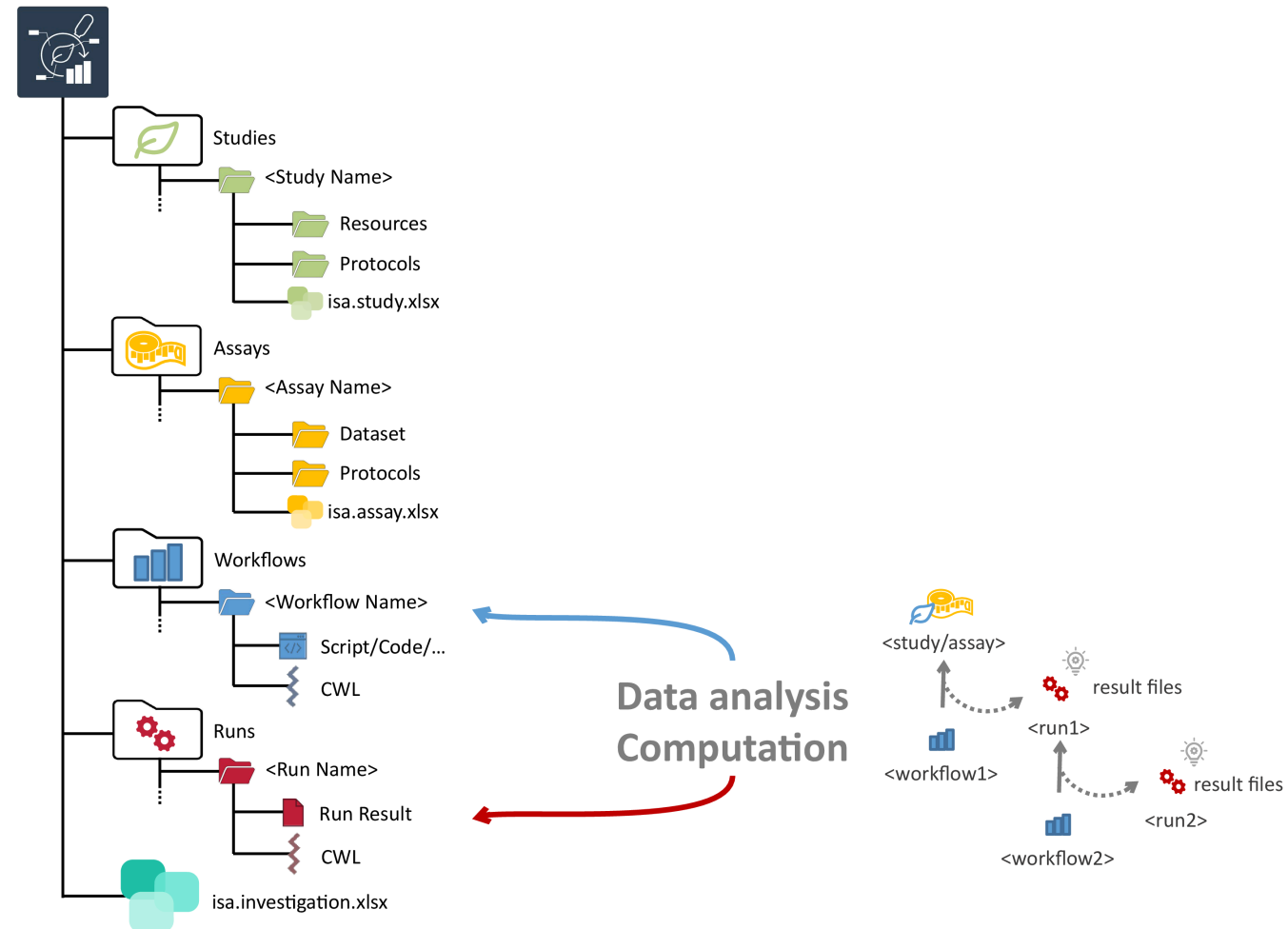
What does an ARC look like?



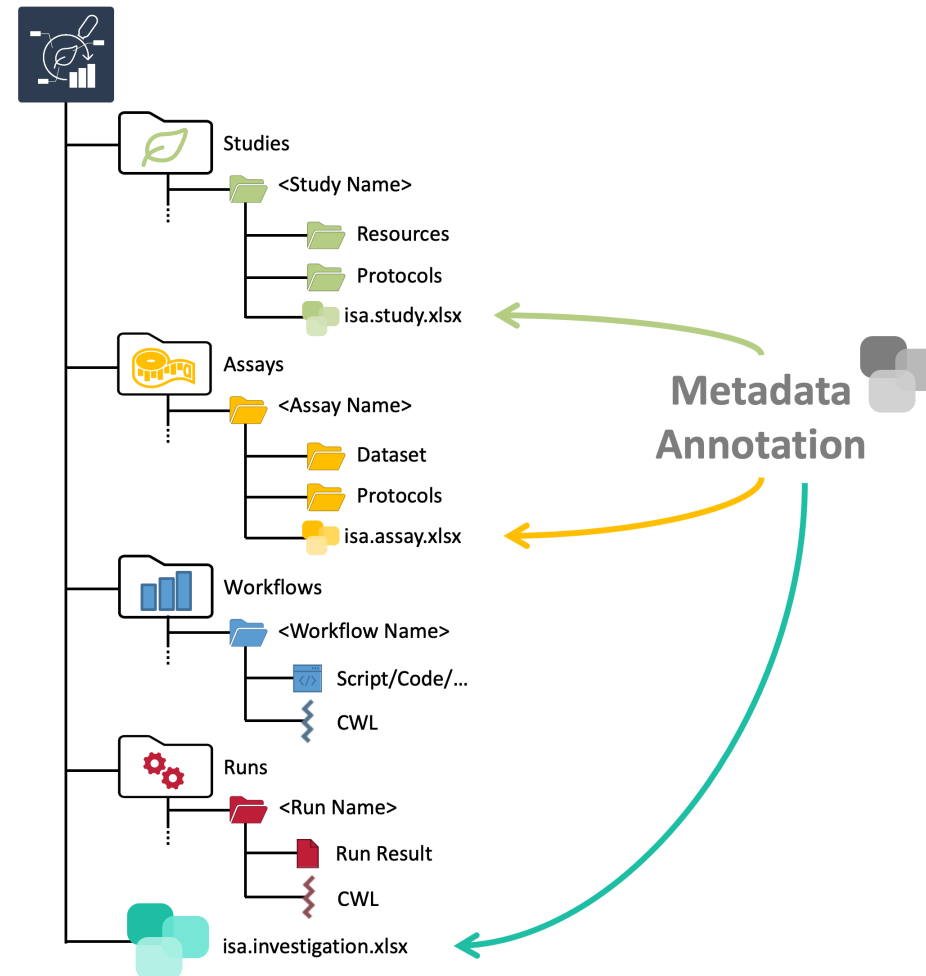
ARCs store experimental data



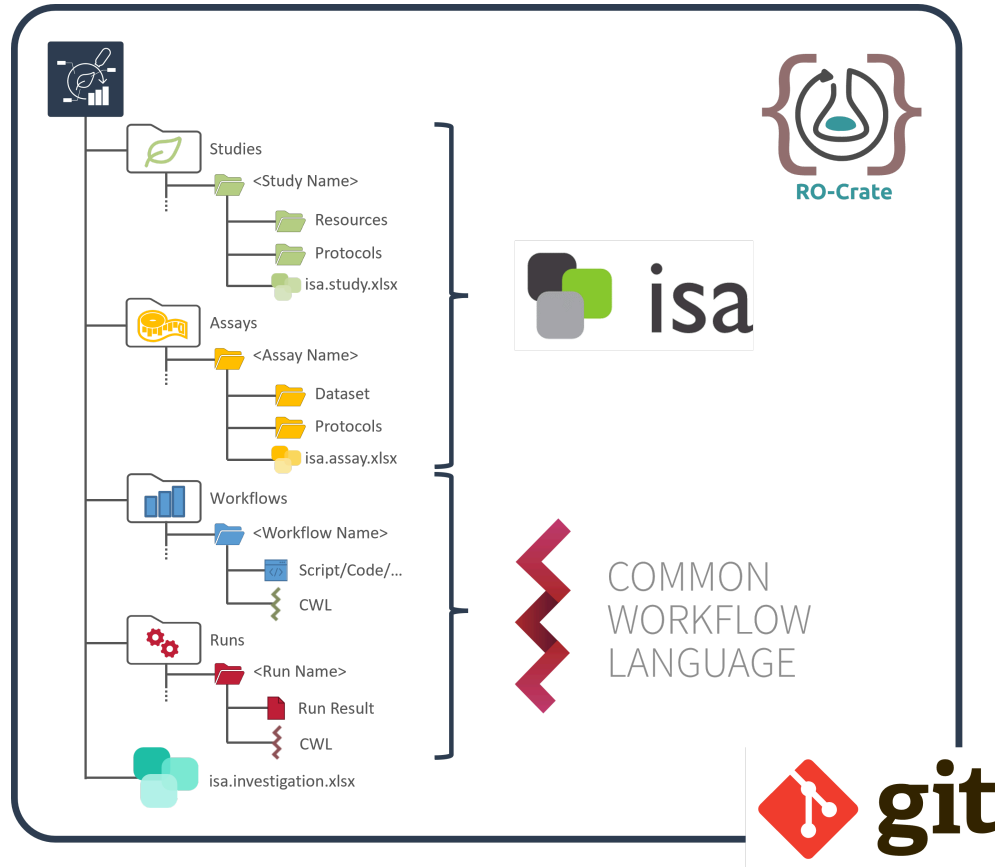
Computations can be run inside ARCs



ARCs come with comprehensive metadata



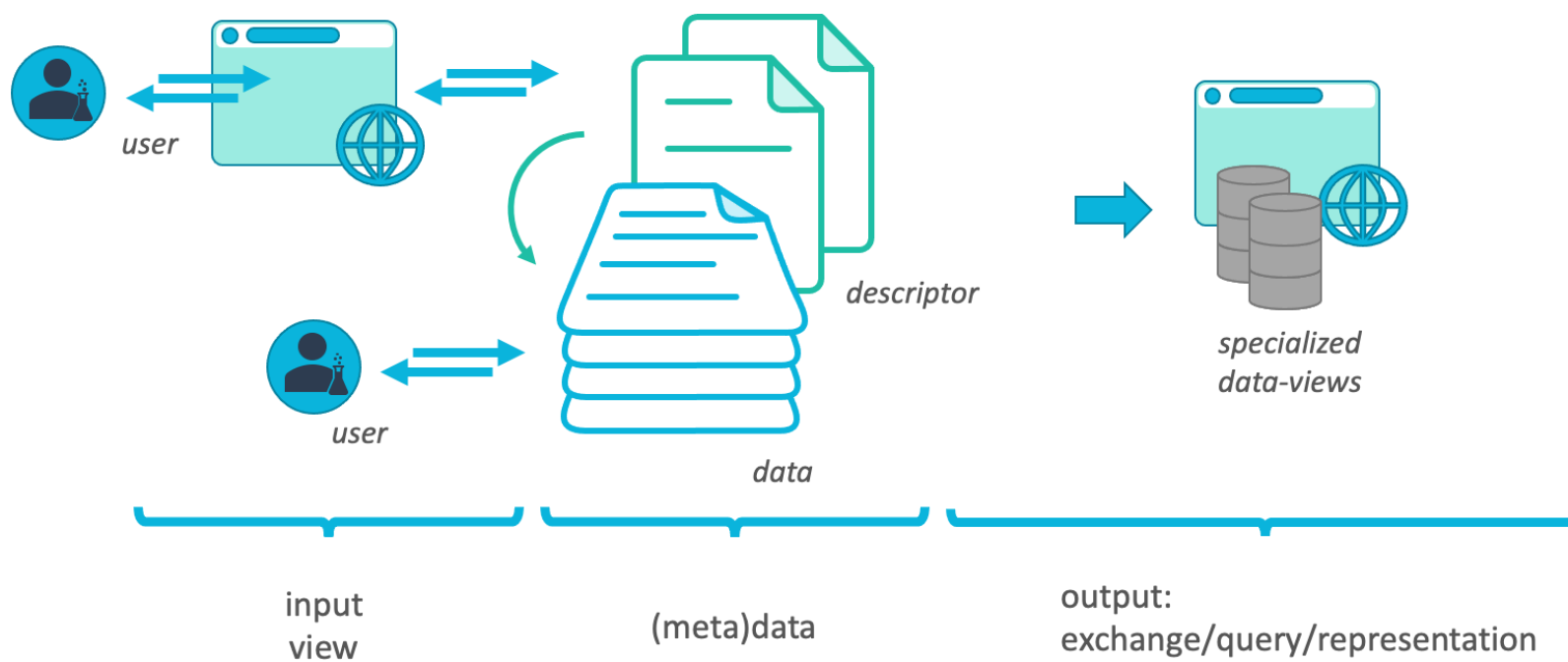
ARC builds on standards



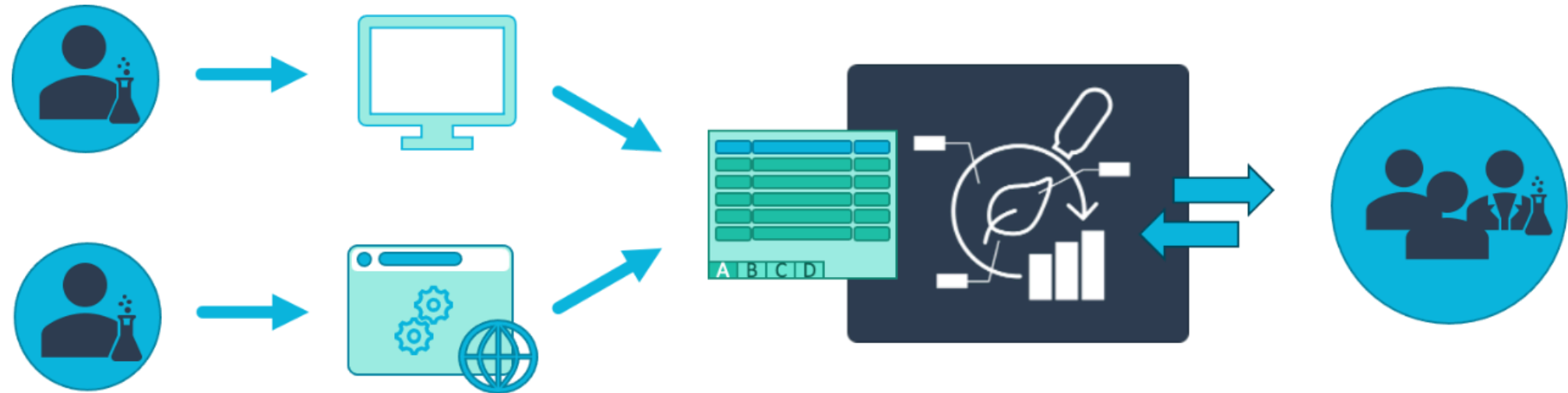
- RO-Crate: standardized exchange
- ISA: structured, machine-readable metadata
- CWL: reproducible, re-usable data analysis
- Git: version control

Everything is a file

DataPLANT employs a **data-centric** approach to RDM



No technical lock-in



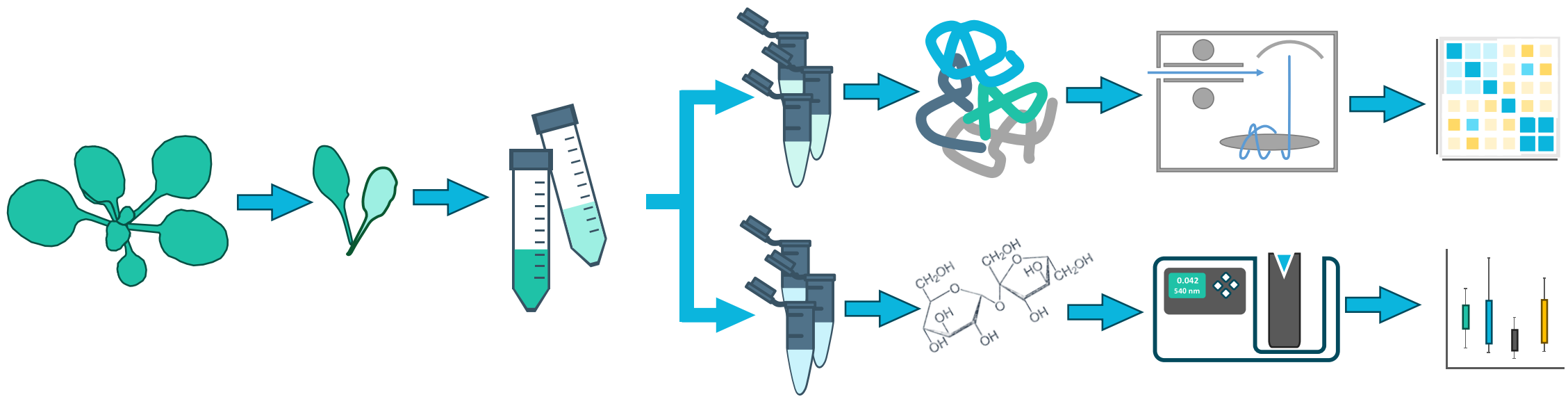
(Meta)data transparency with tool assistance but **no technical lock-in**

Hands-on part 1: Start Here guide

[Start Here guide](#) in the DataPLANT knowledge base.

- follow the fictional researcher Viola on her journey to create her first ARC

A small prototypic project



Hands-on part 1: Setup and Investigation

Follow the [Start Here guide](#) in the DataPLANT knowledge base.

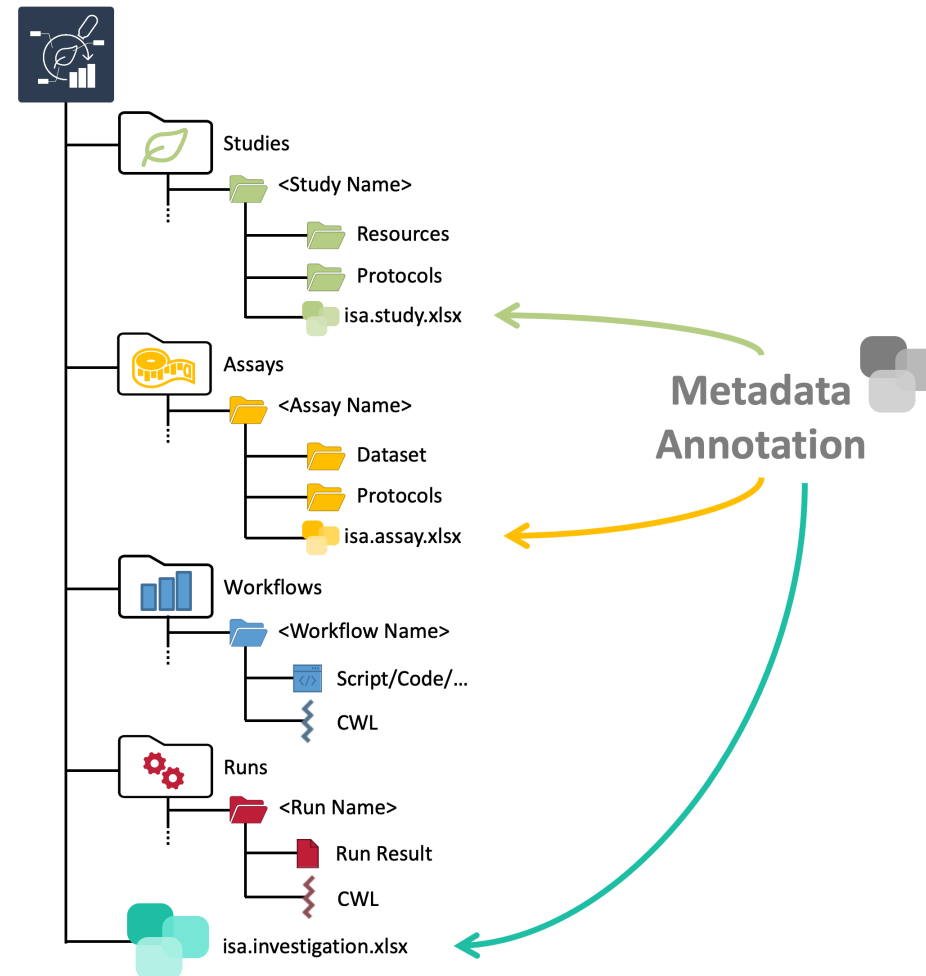
 Stop after step **Investigation**

 Please ask if you have any question/problems 

Metadata

What is metadata?

ARCs come with comprehensive metadata



ISA abstract model in a nutshell



isa

Investigation

administrative (meta)data

- Summary
 - Titel
 - Description
- Person
- Organisation
- Publication reference

Study

descriptive (meta)data
information on the subject

- Characteristics
- Parameters
- Components
- Factors

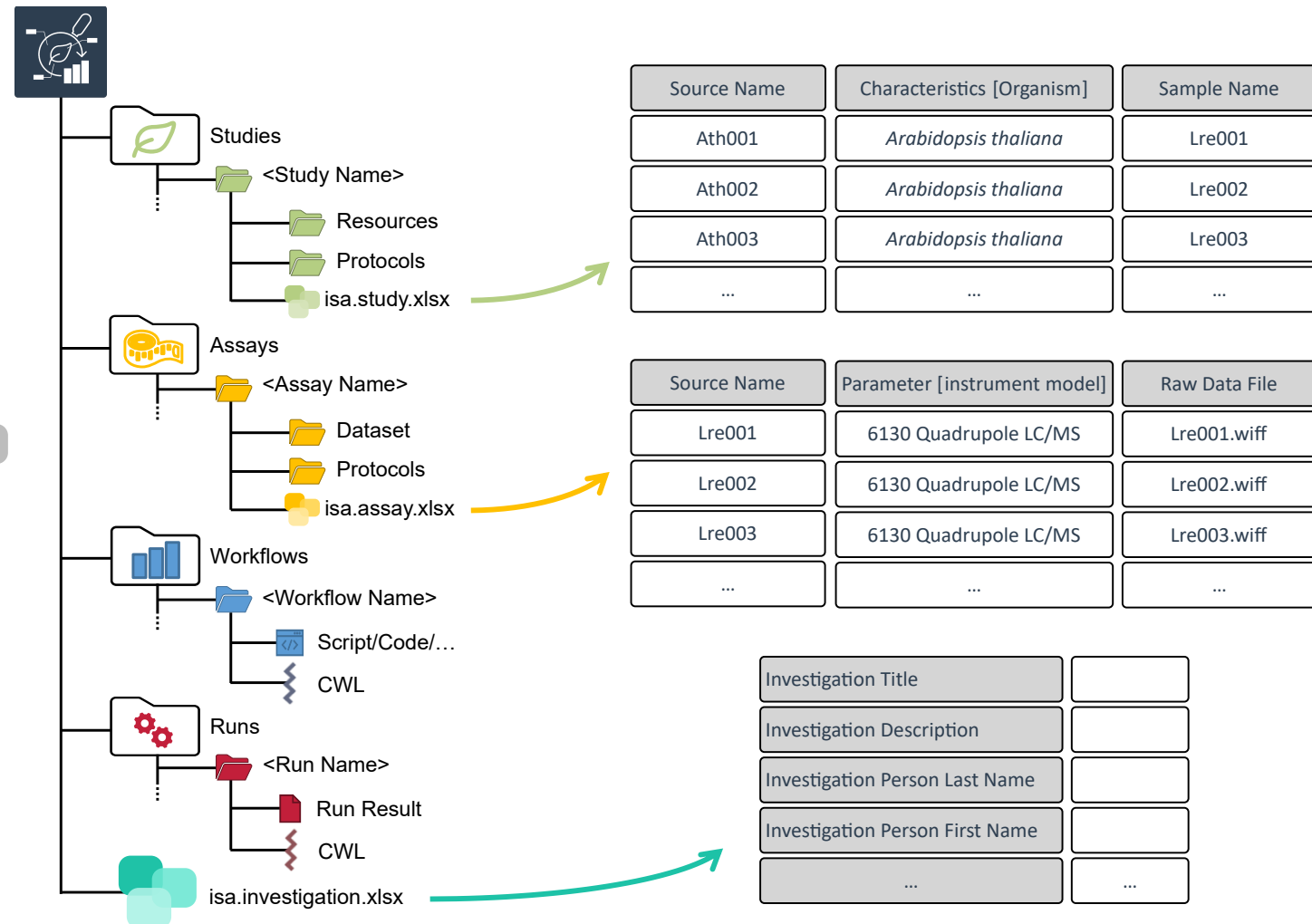
Assay

descriptive (meta)data
information on the measurement

- Characteristics
- Parameters
- Components
- Factors

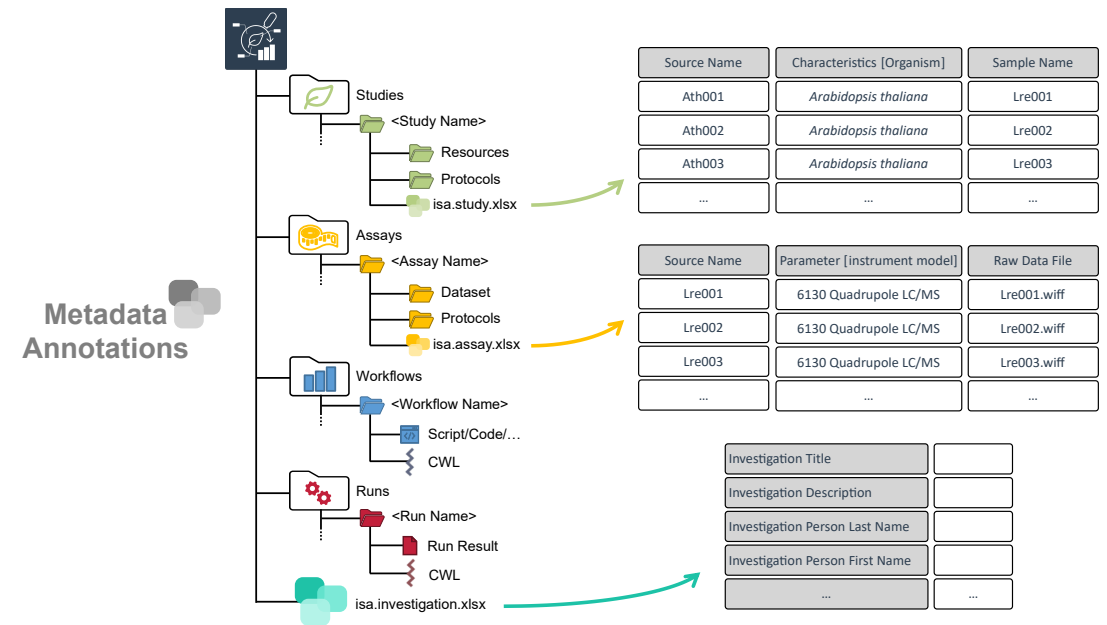
ARC builds on ISA to connect data

Metadata Annotations

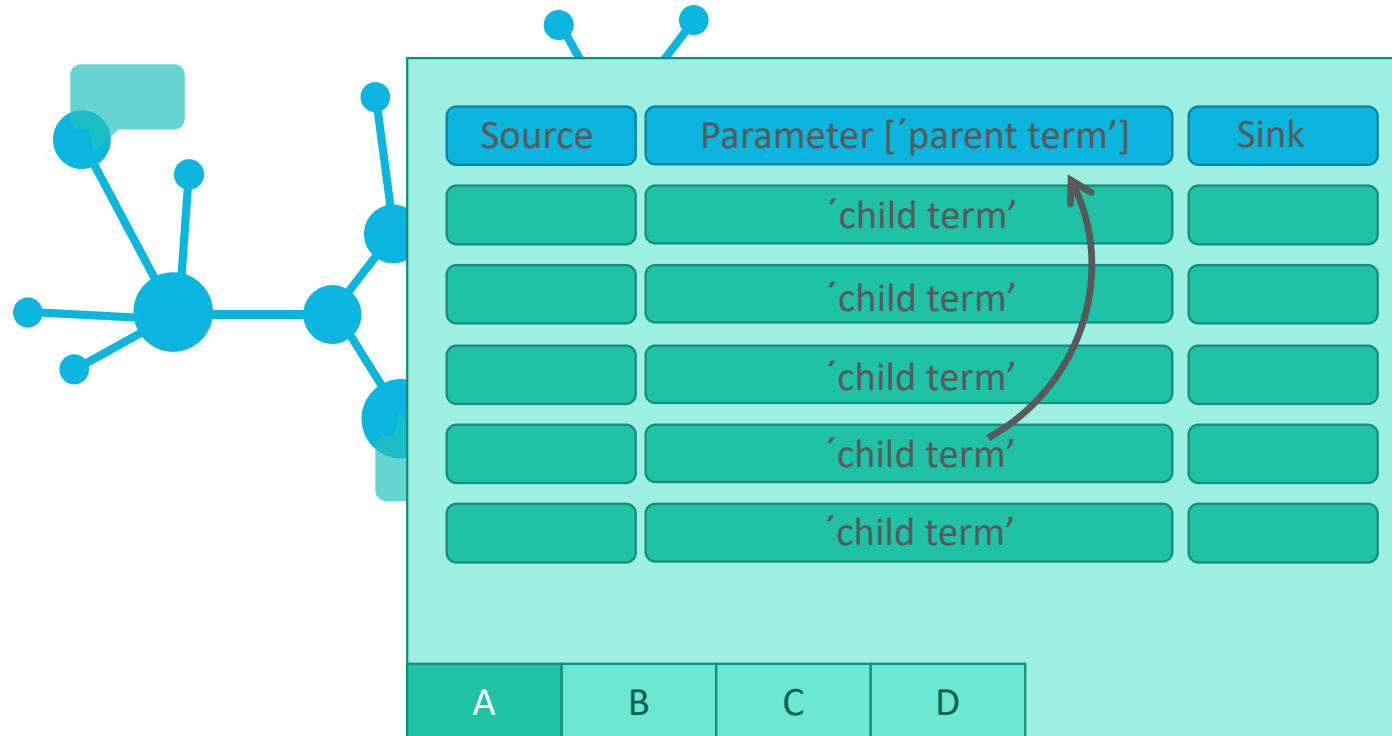


ARC builds on ISA to link data

- Samples are linked study-to-assay, assay-to-assay
- Raw data is linked to assays
- Protocols can be referenced
- ...



Annotation by flattening the knowledge graph



- Low-friction metadata annotation
- Familiar spreadsheet, row/column-based environment

Annotation principle

Sample	Parameter [instrument model]	Data
	'TripleTOF4600 '	
	'TripleTOF4600 '	
	'TripleTOF4600 '	
	'TripleTOF4600 '	
	'TripleTOF4600 '	

A	B	C	D
---	---	---	---

- Low-friction metadata annotation
- Familiar spreadsheet, row/column-based environment

Adding new building blocks

The screenshot shows the Swate interface. On the left is a table with 6 rows and 5 columns: Input [Source Name], Characteristic [organism], Factor [watering exposure], and Output [Sample Name]. The table contains data for various samples (DB_097, DB_099, DB_103, DB_161, DB_163, DB_165) and their corresponding characteristics, factors, and outputs. To the right of the table is a 'New building blocks' dialog. The dialog has a search bar with 'instrument model' entered. Below the search bar is a list of categories: Input, Parameter, Factor, Characteristic, Component, More, and Output. The 'Input' category is selected, and a green 'Add Column' button is visible. Below the dialog is a small input field with the number '1' and a plus sign.

	Input [Source Name]	Characteristic [organism]	Factor [watering exposure]	Output [Sample Name]
1	DB_097	Talinum fruticosum ✓	12 days drought	CAM_01
2	DB_099	Talinum fruticosum ✓	12 days drought	CAM_02
3	DB_103	Talinum fruticosum ✓	12 days drought	CAM_03
4	DB_161	Talinum fruticosum ✓	12 days drought + 2 days rewatered	reC3_01
5	DB_163	Talinum fruticosum ✓	12 days drought + 2 days rewatered	reC3_02
6	DB_165	Talinum fruticosum ✓	12 days drought + 2 days rewatered	reC3_03

Metadata +

Swate (now integrated in ARCitect) can be used for the annotation of **isa.study.xlsx** and **isa.assay.xlsx** files

Annotation Building Block types

- Input (e.g. Source Name, Sample Name)
- Protocol columns
- Characteristic // Parameter // Factor // Component
- Output (e.g. Sample Name, Raw Data File, Derived Data File)

The screenshot displays a software interface for building annotations. It features a table with four main columns: 'Input [Source Name]', 'Characteristic [organism]', 'Factor [watering exposure]', and 'Output [Sample Name]'. The table lists six entries, each with a unique ID, a source name, an organism (Talinum fruticosum), a factor (e.g., 12 days drought), and an output (e.g., CAM_01). A 'New Parameter' dialog is open on the right, allowing users to search for and add new parameters. The dialog includes a search bar and a list of categories: Input, Parameter, Factor, Characteristic, Component, More, and Output. A 'Sidebar' label points to the right edge of the interface.

	Input [Source Name]	Characteristic [organism]	Factor [watering exposure]	Output [Sample Name]
1	DB_097	Talinum fruticosum	✓ 12 days drought	CAM_01
2	DB_099	Talinum fruticosum	✓ 12 days drought	CAM_02
3	DB_103	Talinum fruticosum	✓ 12 days drought	CAM_03
4	DB_161	Talinum fruticosum	✓ 12 days drought + 2 days rewatered	reC3_01
5	DB_163	Talinum fruticosum	✓ 12 days drought + 2 days rewatered	reC3_02
6	DB_165	Talinum fruticosum	✓ 12 days drought + 2 days rewatered	reC3_03

Let's take a detour on [Annotation Principles](#)

Ontology term search

The screenshot displays the Swate application interface. On the left, a table with four columns: 'Input [Source Name]', 'Characteristic [Organism]', 'Factor [watering exposure]', and 'Output [Sample Name]'. The table contains six rows of data. Below the table is a pagination control showing '1' and a '+' button. At the bottom of the table area is a metadata bar with 'Metadata', 'plant_material', and a '+' button. On the right, a sidebar titled 'Ontology term search' contains a search input field, a 'Parent: Organism, MIAPPE:0041' label with a 'Use advanced search' link, and a 'Fill selected cells with this term' button. The Swate logo is in the top right of the sidebar, and the version 'Swate Release Version v1.0.0-beta.03 Host Browser' is at the bottom.

	Input [Source Name]	Characteristic [Organism]	Factor [watering exposure]	Output [Sample Name]
1	DB_097		12 days drought	CAM_01
2	DB_099		12 days drought	CAM_02
3	DB_103		12 days drought	CAM_03
4	DB_161		12 days drought + 2 days rewatered	reC3_01
5	DB_163		12 days drought + 2 days rewatered	reC3_02
6	DB_165		12 days drought + 2 days rewatered	reC3_03

1 +

Metadata plant_material +

Swate Release Version v1.0.0-beta.03 Host Browser

Ontology term search

Search for an ontology term to fill into the selected field(s)

Parent: Organism, MIAPPE:0041 [Use advanced search](#)

Fill selected cells with this term

Enable **related term directed search** to directly fill cells with child terms

Fill your table with ontology terms

	Input [Source Name]	Characteristic [Organism]	Factor [watering exposure]	Output [Sample Name]
1	DB_097	Talinum fruticosum ✓	12 days drought	CAM_01
2	DB_099	Talinum fruticosum ✓	12 days drought	CAM_02
3	DB_103	Talinum fruticosum ✓	12 days drought	CAM_03
4	DB_161	Talinum fruticosum ✓	12 days drought + 2 days rewatered	reC3_01
5	DB_163	Talinum fruticosum ✓	12 days drought + 2 days rewatered	reC3_02
6	DB_165	Talinum fruticosum ✓	12 days drought + 2 days rewatered	reC3_03

1 +

Metadata plant_material +

SWATE

Ontology term search

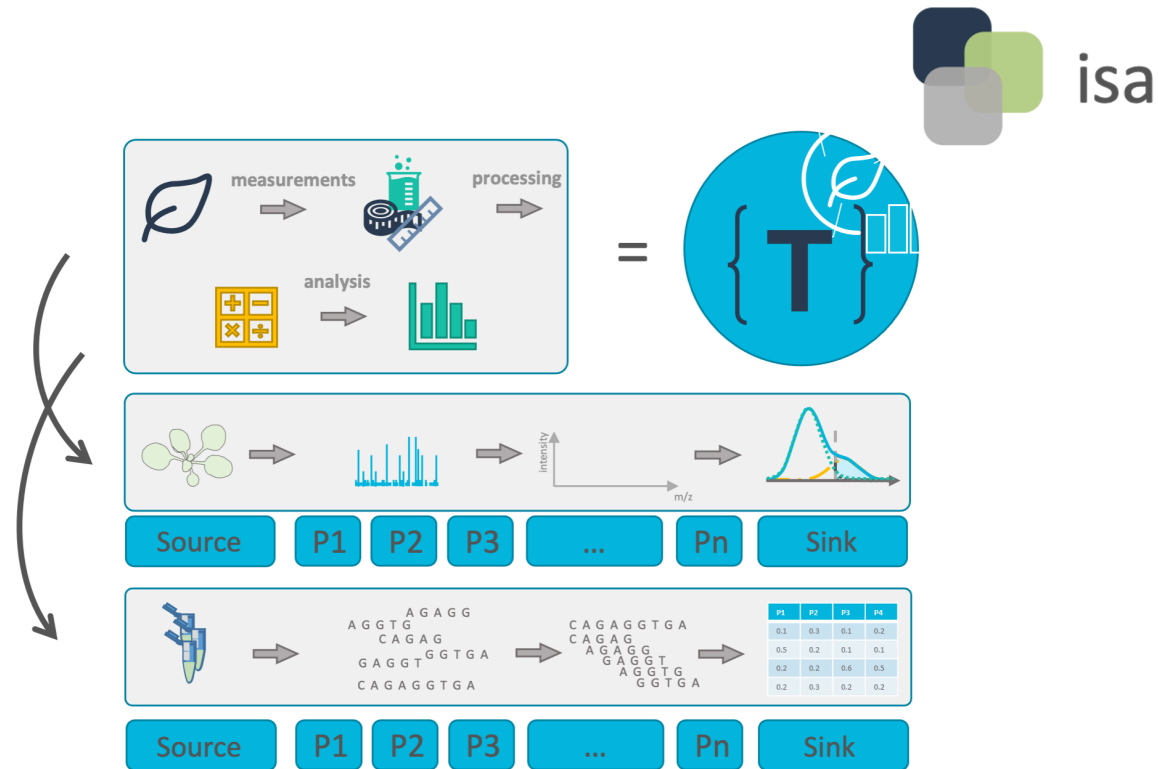
Search for an ontology term to fill into the selected field(s)

Parent: Organism, MIAPPE:0041 [Use advanced search](#)

Fill selected cells with this term

Swate Release Version v1.0.0-beta.03 Host [Browser](#)

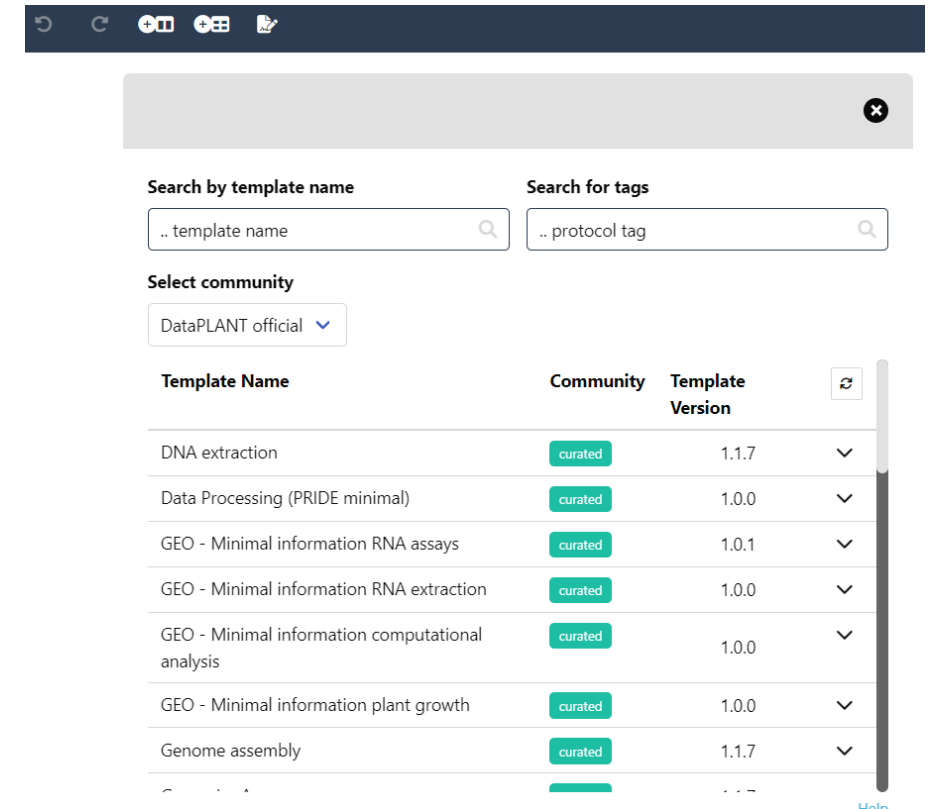
Realization of lab-specific metadata templates



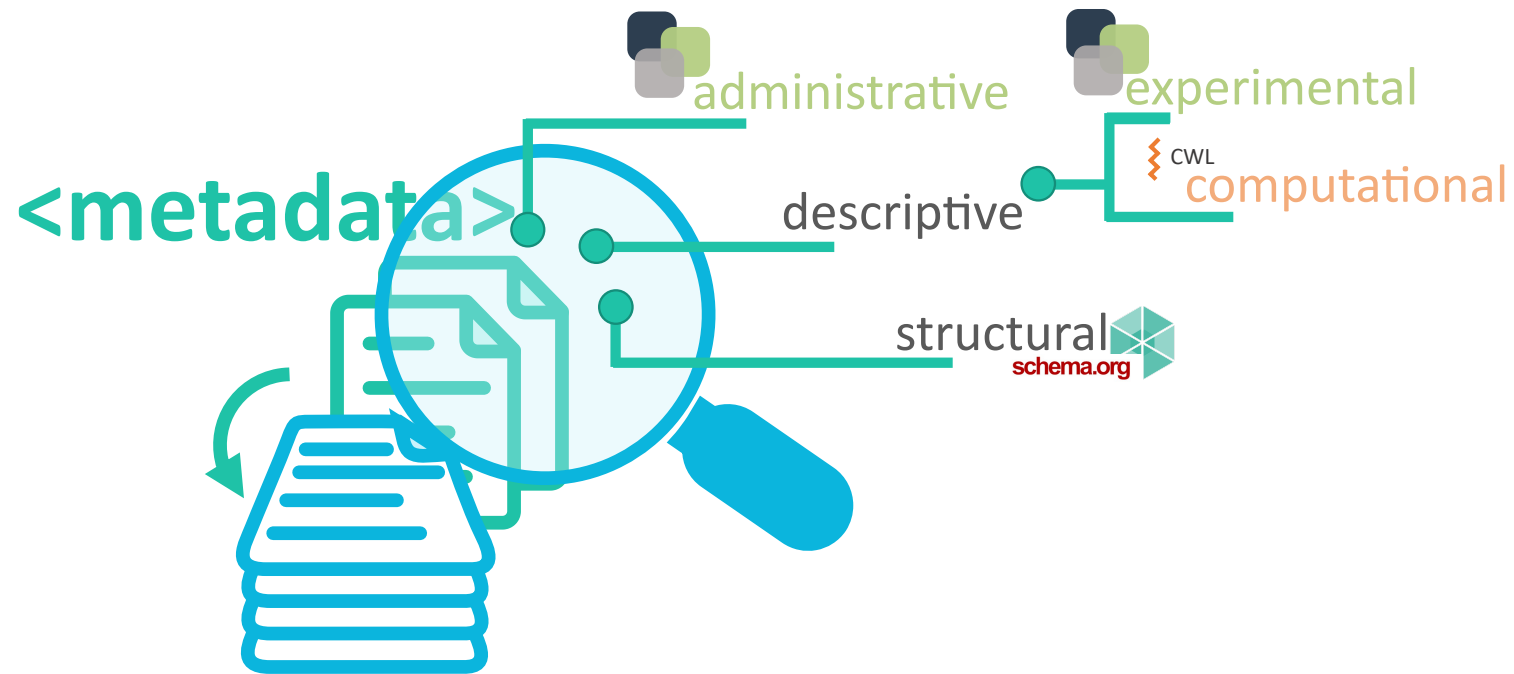
Facilities can define their most common workflows as templates

Directly import templates via Swate

- DataPLANT curated
- Community templates

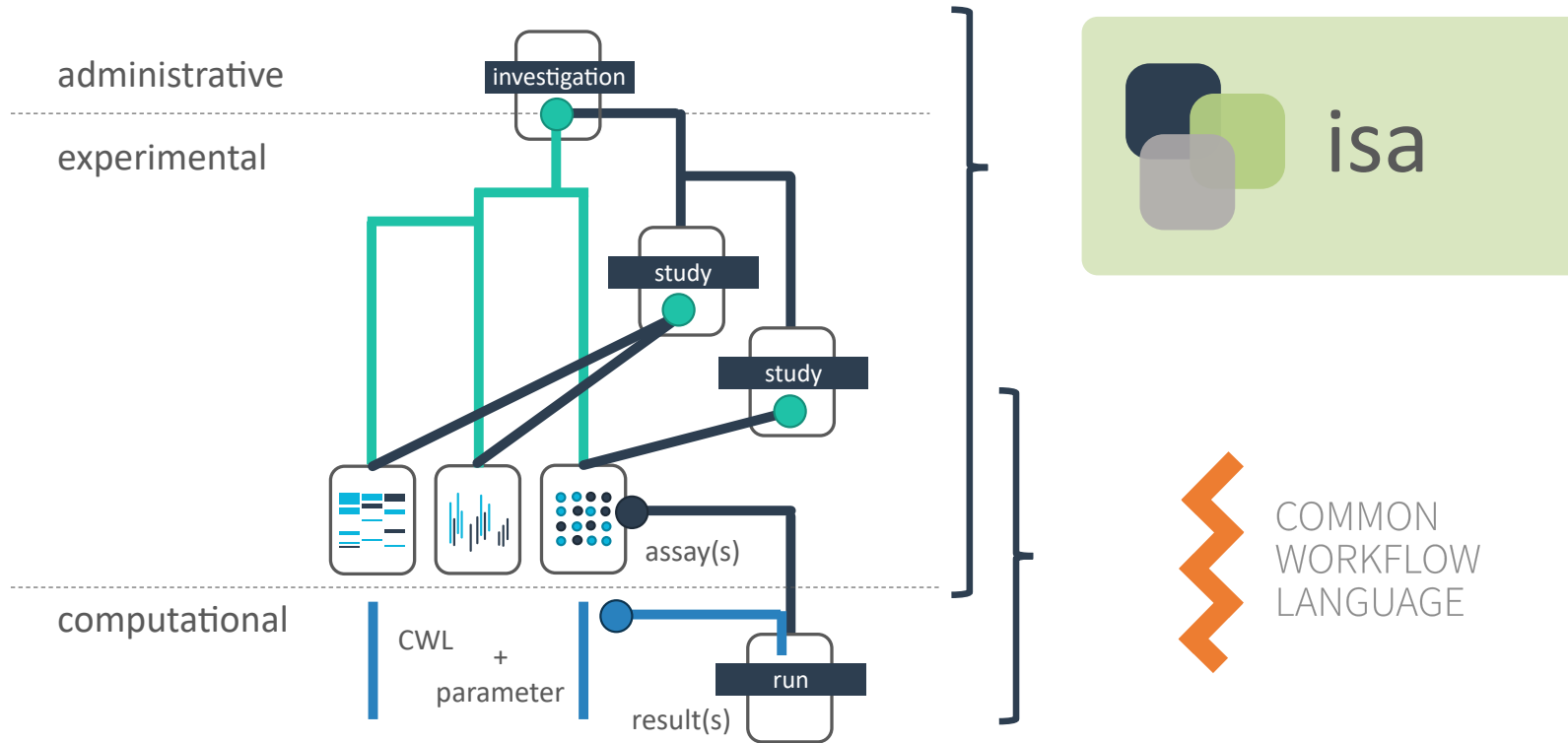


Build on existing well-established standards



Integration of established (meta)data standards ensures compatibility by design

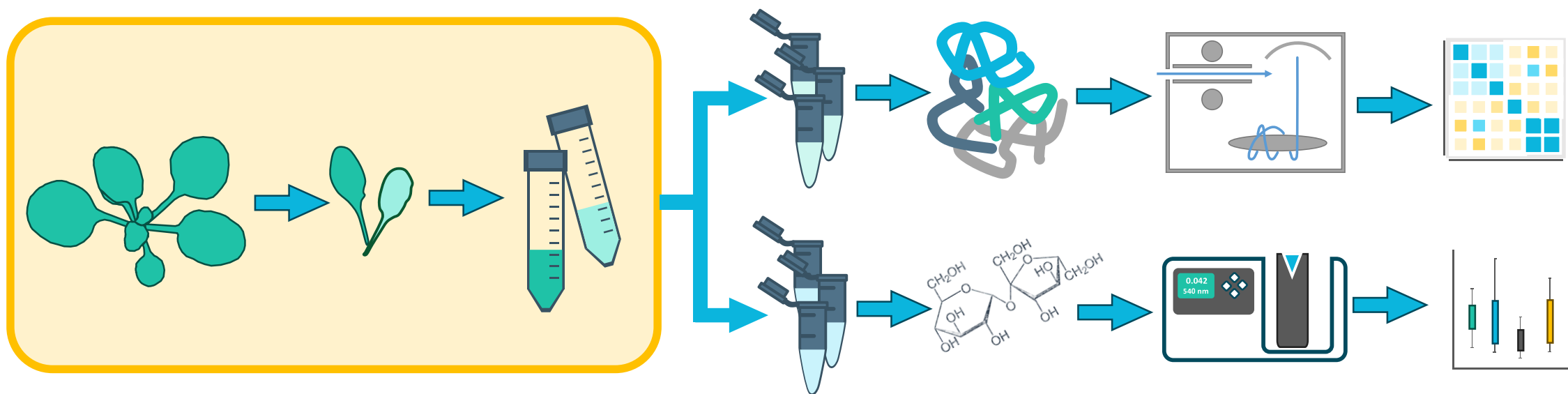
ISA and CWL – Connected by similarity



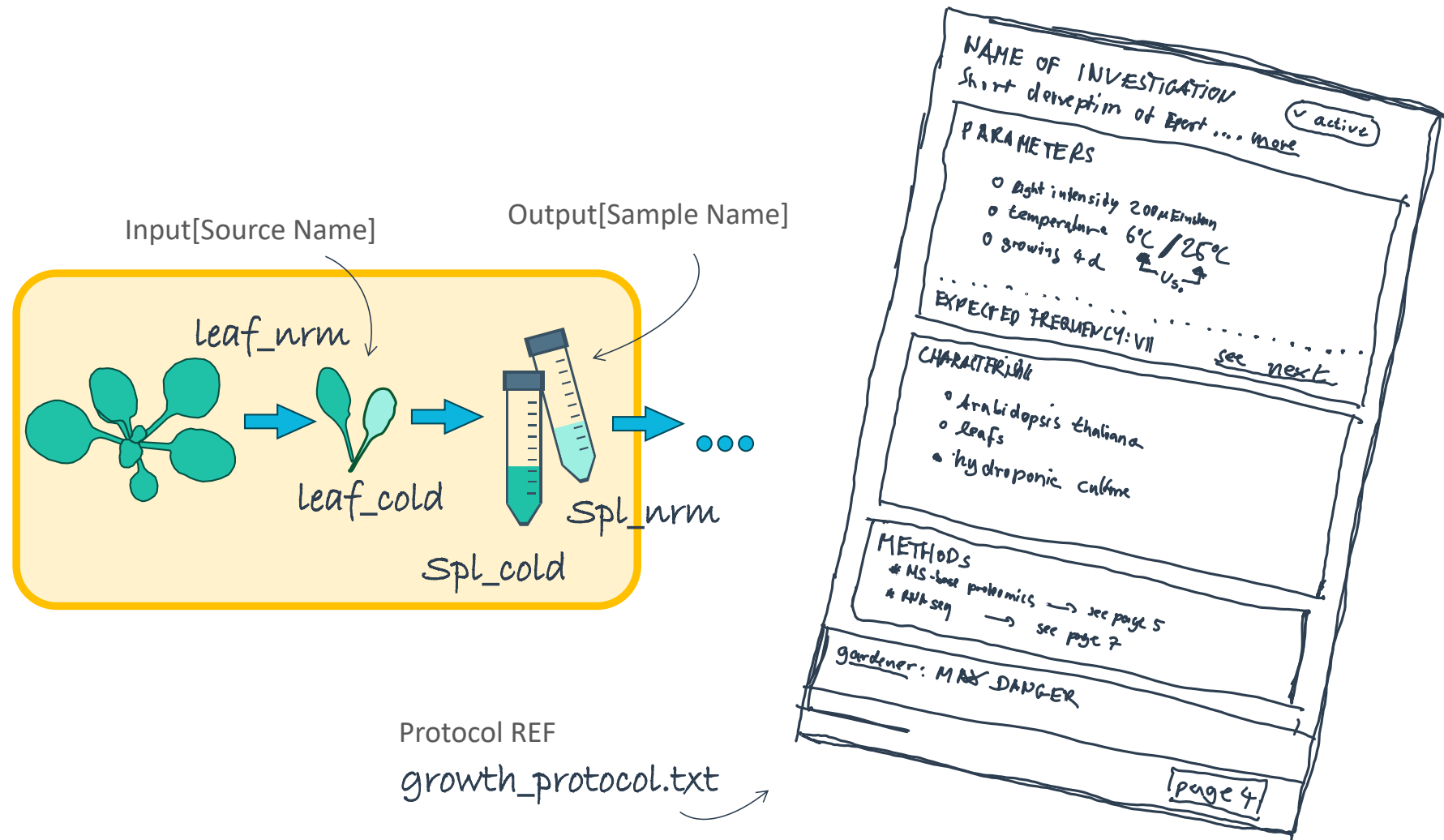
Hands-on part 2: Start Here guide, study & assay part

[Start Here guide](#) in the DataPLANT knowledge base.

Divide and conquer for reproducibility



Identifying the 'study' part



A table-based organization schema

Input[Source Name]					Output[Sample Name]			
leaf_nrm					spl_nrm			
leaf_cold					spl_cold			
A	B	C	D					

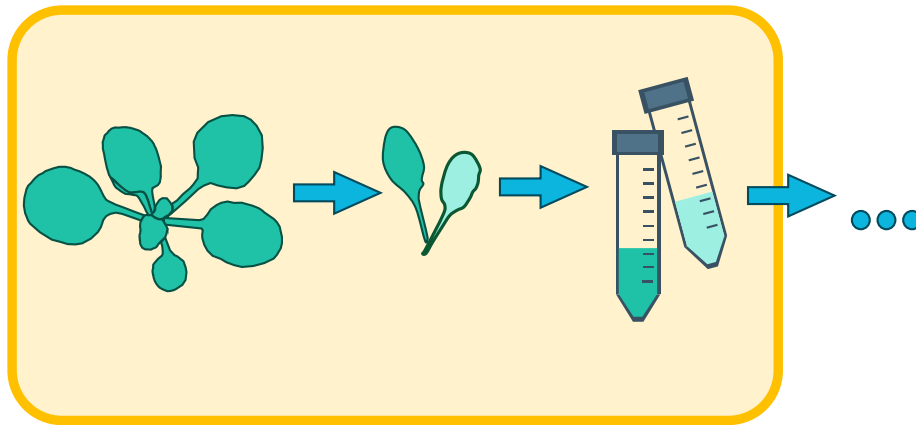
Referencing a protocol

This allows you to reference the free-text, human-readable protocol.

Input[Source Name]		Protocol REF	Output[Sample Name]	
leaf_nrm		growth_protocol.txt	spl_nrm	
leaf_cold		growth_protocol.txt	spl_cold	
A	B	C	D	

- 💡 It is recommended that the protocol is in an open format (.md|.txt|.docx|...)
- 💡 But everything is possible also an URI to an electronic lab notebook

Parameterizing the 'study'



NAME OF INVESTIGATION
Short description of Experiment... more active

PARAMETERS

- o Light intensity 200 μ Einstein
- o temperature 6°C / 25°C
- o growing 4 d $\uparrow$ $\downarrow$ U_s

EXPECTED FREQUENCY: VII see next

CHARACTERISTICS

- o *Arabidopsis thaliana*
- o leaves
- o hydroponic culture

METHODS

- * MS-based proteomics $\rightarrow$ see page 5
- * RNA seq $\rightarrow$ see page 7

Gardener: MAX DANGER

page 4

Finding the right metadata vocabulary

Parameters []

- Light intensity 200 μ Einstein
- Temperature 6°C / 25°C
- Growing 4d

Characteristics []

- *Arabidopsis thaliana*
- Leaf
- Hydroponic culture
- Columbia

OLS: Finding the right metadata vocabulary

Temperature Dependence Temperature:Dependence_Annotation

http://purl.uniprot.org/core/Temperature_Dependence_Annotation

Indicates the optimum temperature for enzyme activity and/or the variation of enzyme activity with temperature variation; the thermostability/thermolability of the enzyme is also mentioned when it is known.

Ontology: UNIPROTDFS

temperature AFO:/result#AFR_0001584

http://purl.allotrope.org/ontologies/result#AFR_0001584

A temperature (datum) is a quantity facet that quantifies some temperature. [Allotrope]

Ontology: AFO

temperature FBcv:0000466

http://purl.obolibrary.org/obo/FBcv_0000466

Mutation caused by exposure to a temperature that is higher or lower than 25 degrees Celsius.

Ontology: FBCV

temperature PATO:0000146

http://purl.obolibrary.org/obo/PATO_0000146

A physical quality of the thermal energy of a system.

Ontology: PATO

Also appears in: NGBO HTN CAO ZP AGRO OMIABIS OBIB MONDO TXPO MCO +

Welcome to the EMBL-EBI Ontology Lookup Service

temperature

Search

☐ Exact match ☐ Include obsolete terms ☒ Include imported terms

Examples: diabetes, GO:0098743

Looking for a particular ontology?

Data Content
Updated 4 Oct 2024 Fri 17:50(+02:00)

- 265 ontologies
- 8,598,500 classes
- 45,471 properties
- 687,188 individuals

About OLS
The Ontology Lookup Service (OLS) is a repository for biomedical ontologies that aims to provide a single point of access to the latest ontology versions. You can browse the ontologies through the website as well as programmatically via the OLS API. OLS is developed and maintained by the Samples, Phenotypes and Ontologies Team (SPOT) at EMBL-EBI.

Related Tools
In addition to OLS the SPOT team also provides the **OxO** and **ZOOMA** services. OxO provides cross-ontology mappings between terms from different ontologies. ZOOMA is a service to assist in mapping data to ontologies in OLS.

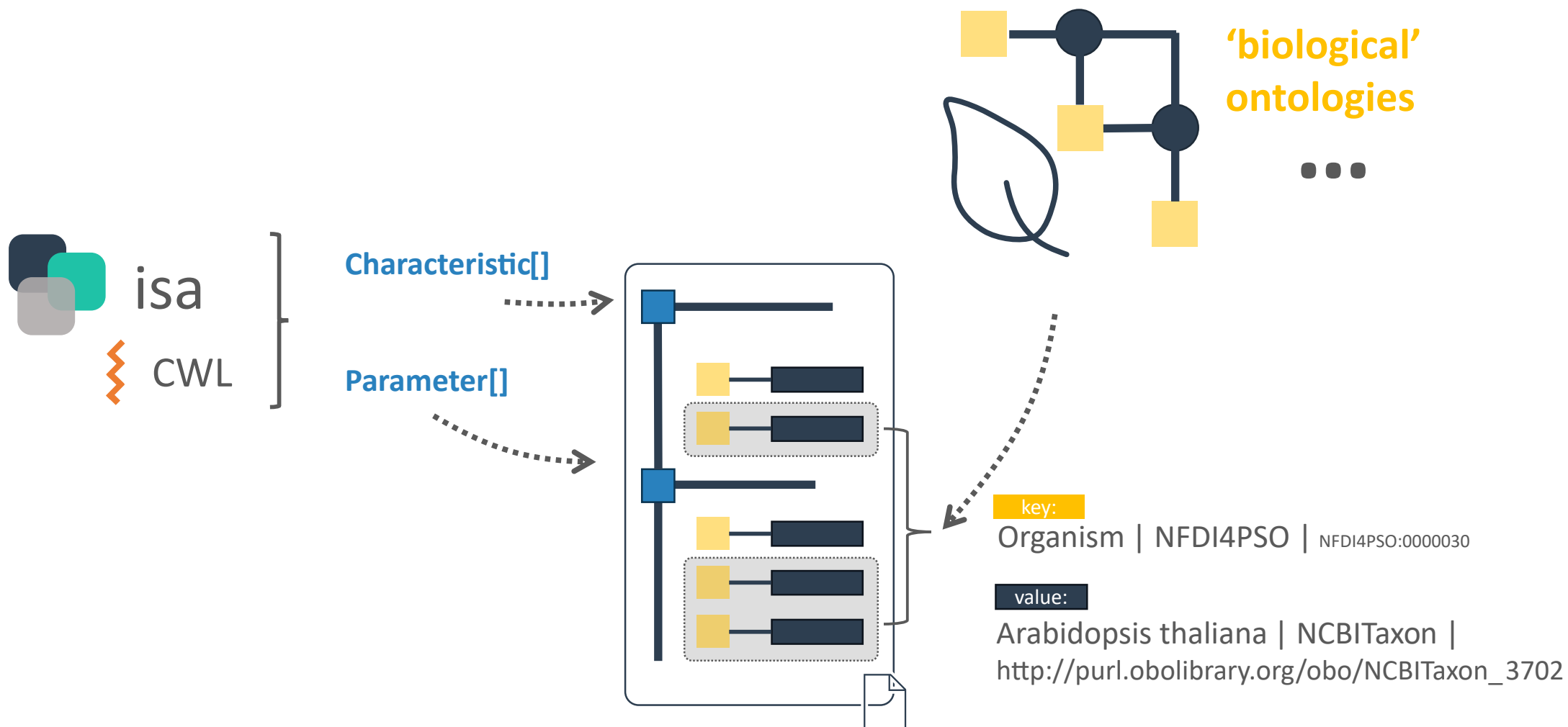
Report an Issue
For feedback, enquiries or suggestion about OLS or to request a new ontology please use our [GitHub issue tracker](#). For announcements relating to OLS, such as new releases and new features sign up to the [OLS announce mailing list](#).

FOLLOW US

EMBL-EBI 2023 Licensing

Ontology Lookup Service (OLS): <https://www.ebi.ac.uk/ols4/>

Finding the metadata vocabulary and descriptors



Finding the metadata vocabulary and descriptors

Parameters []

- 
 -  Light intensity 200 μ Einstein
- 
 -  Temperature 6°C / 25°C
- 
 -  Growing 4d

Characteristics []

- 
 -  *Arabidopsis thaliana*
- 
 -  Leaf
- 
 -  Hydroponic culture
- 
 -  Columbia

Finding the metadata vocabulary and descriptors

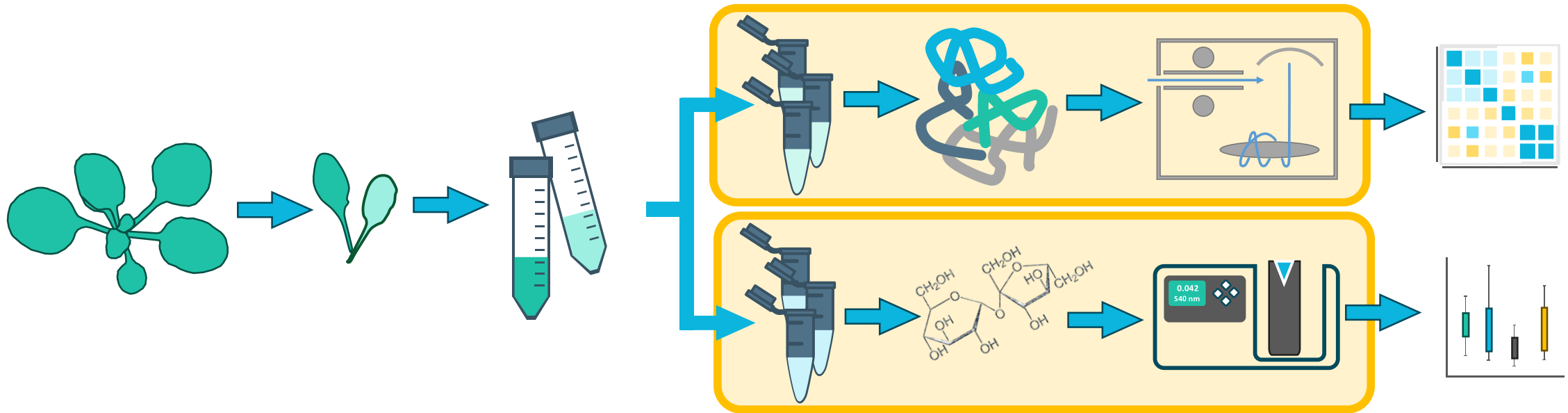
Parameters []

-  Light intensity
 -  200 μ Einstein
-  Temperature
 -  6°C / 25°C
-  Growth time
 -  4d

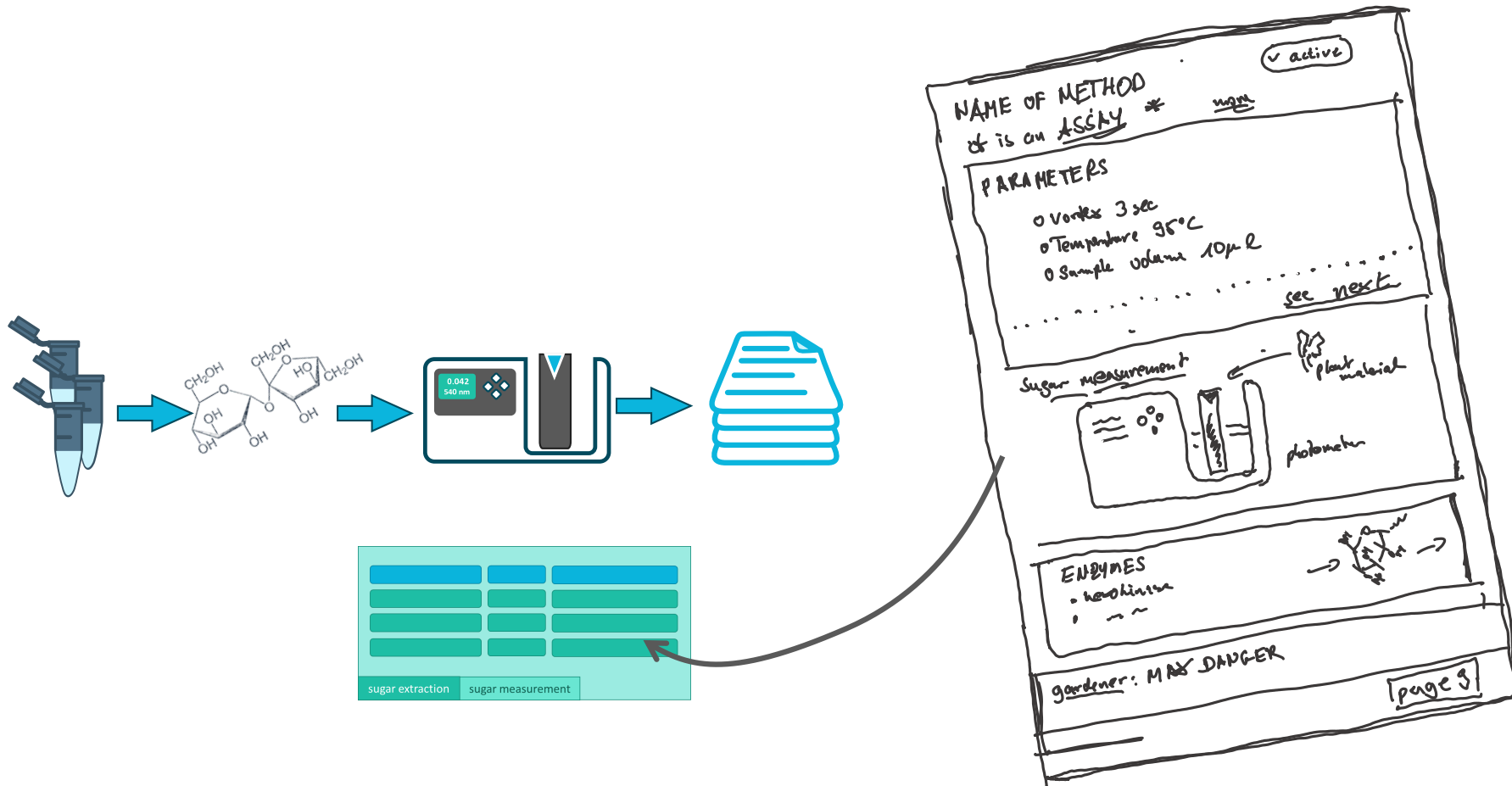
Characteristics []

-  Organism
 -  *Arabidopsis thaliana*
-  Tissue
 -  Leaf
-  Growth medium
 -  Hydroponic culture
-  Ecotype
 -  Columbia

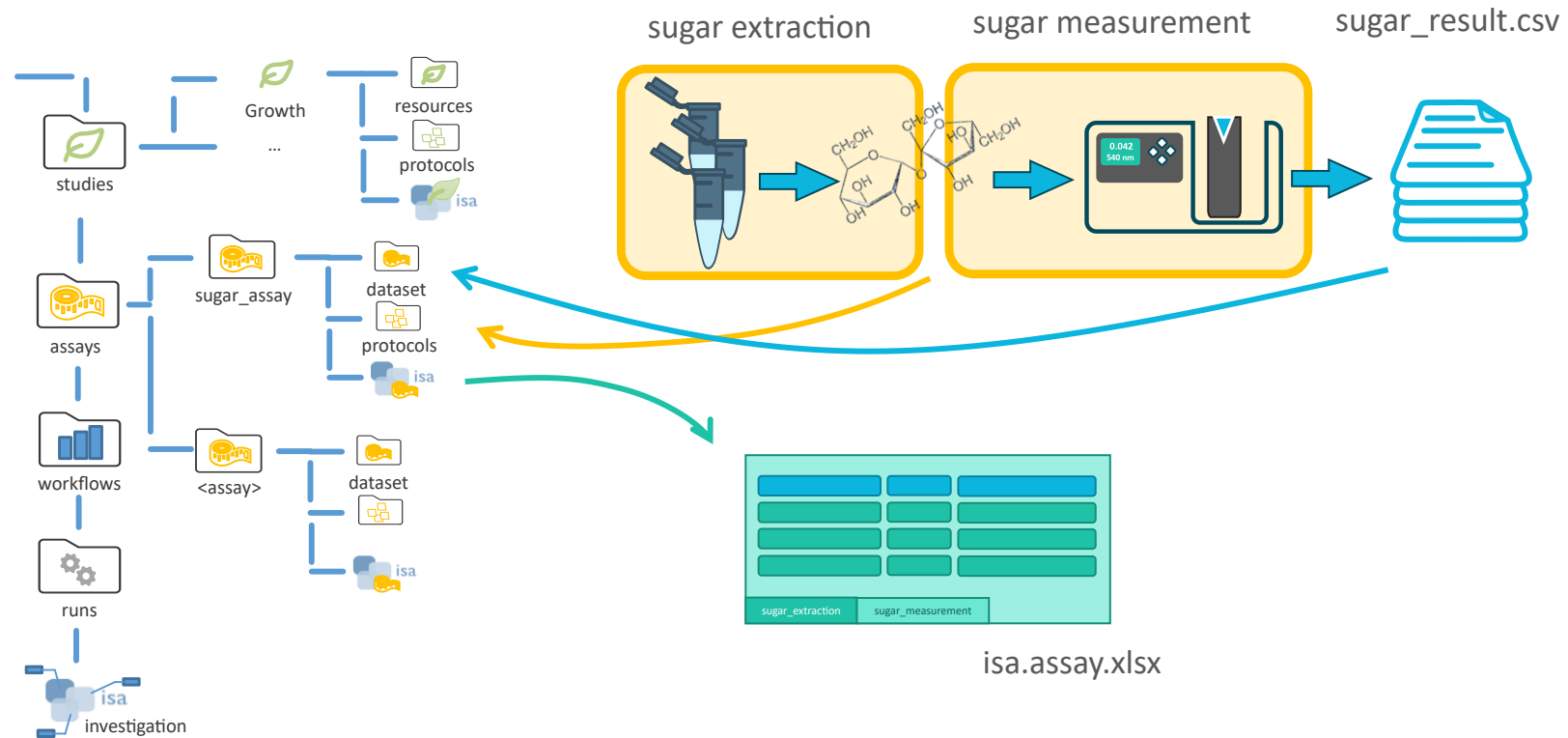
Identifying assays



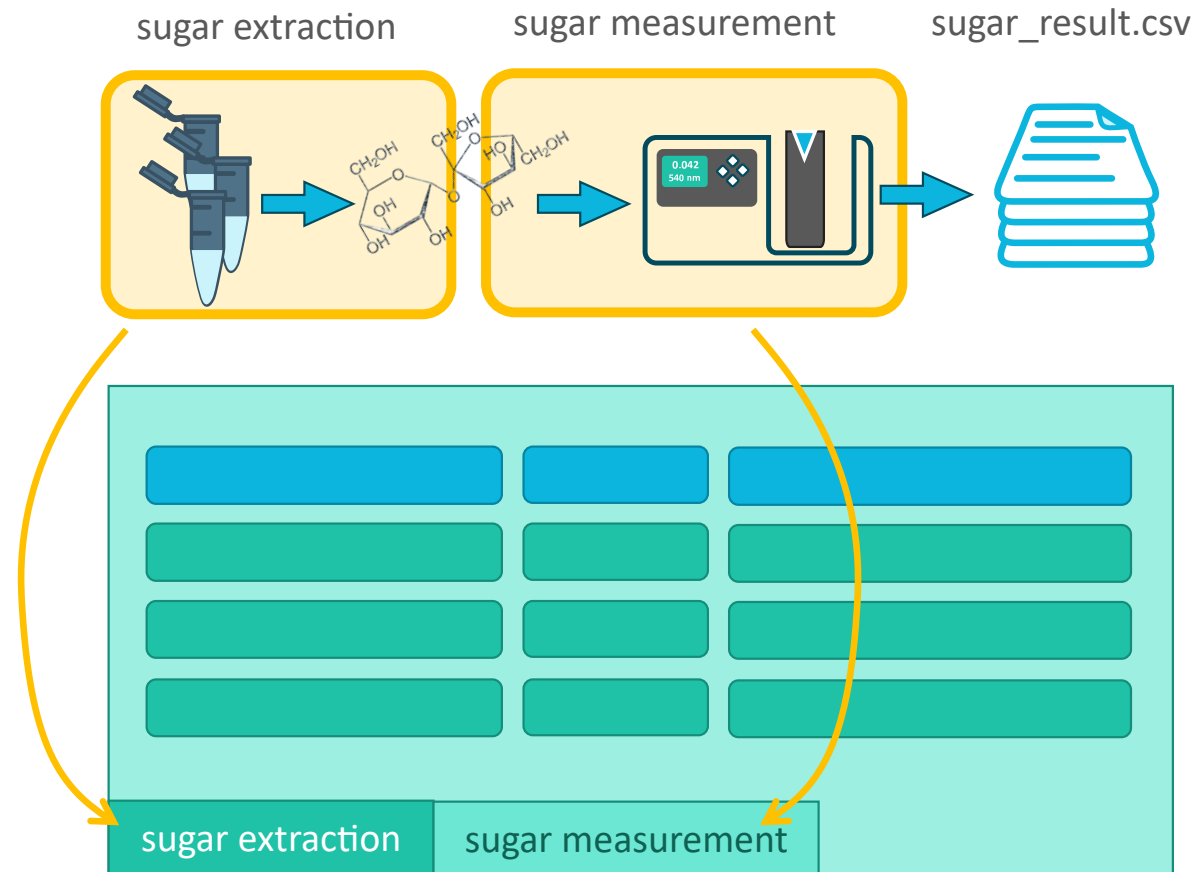
Assay for sugar measurement



Separating different assay elements

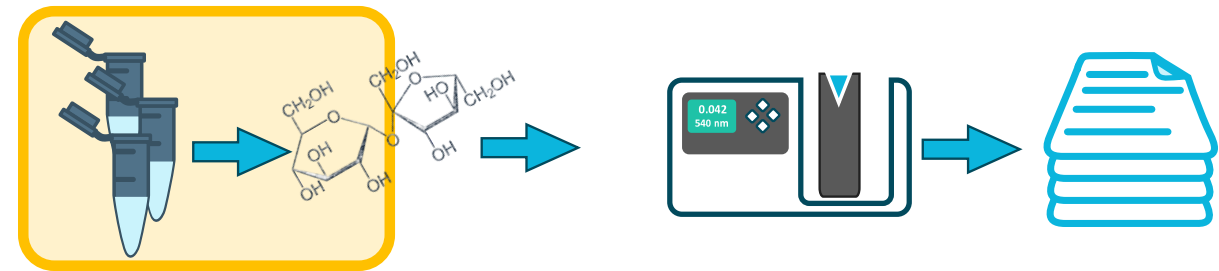


Isolating the lab processes in an assay



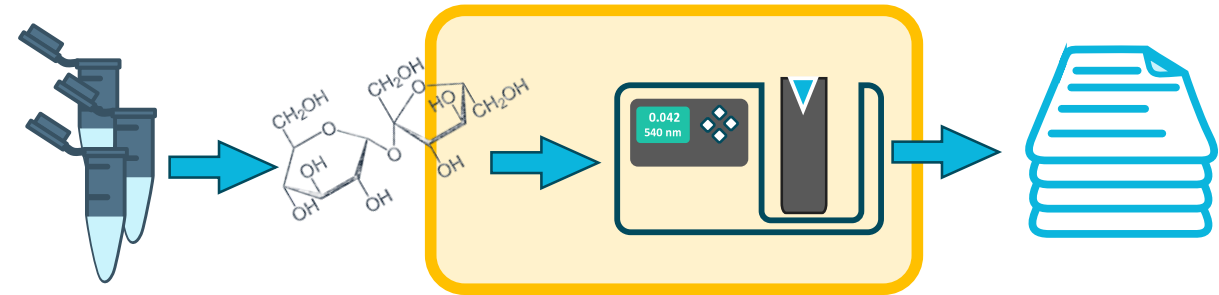
Parameterization: sugar extraction

-  Vortex Mixer
 -  3 seconds
-  Temperature
 -  95 degree celsius



Parameterization: sugar measurement

-  technical replicate
 -  1,2,3,...
-  sample volume
 -  10 microliter
-  buffer volume
 -  190 microliter
-  cycle count
 -  5



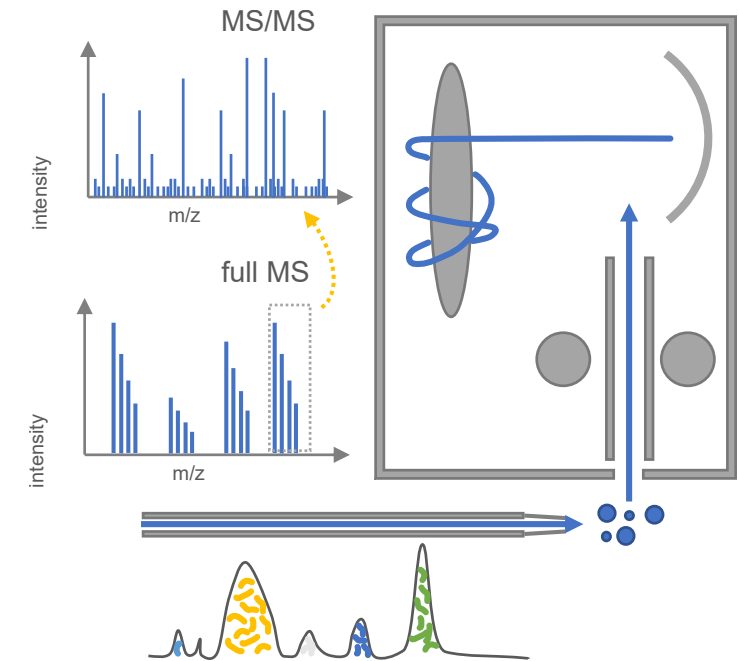
Save time using standard methods and SOPs

Parameter []

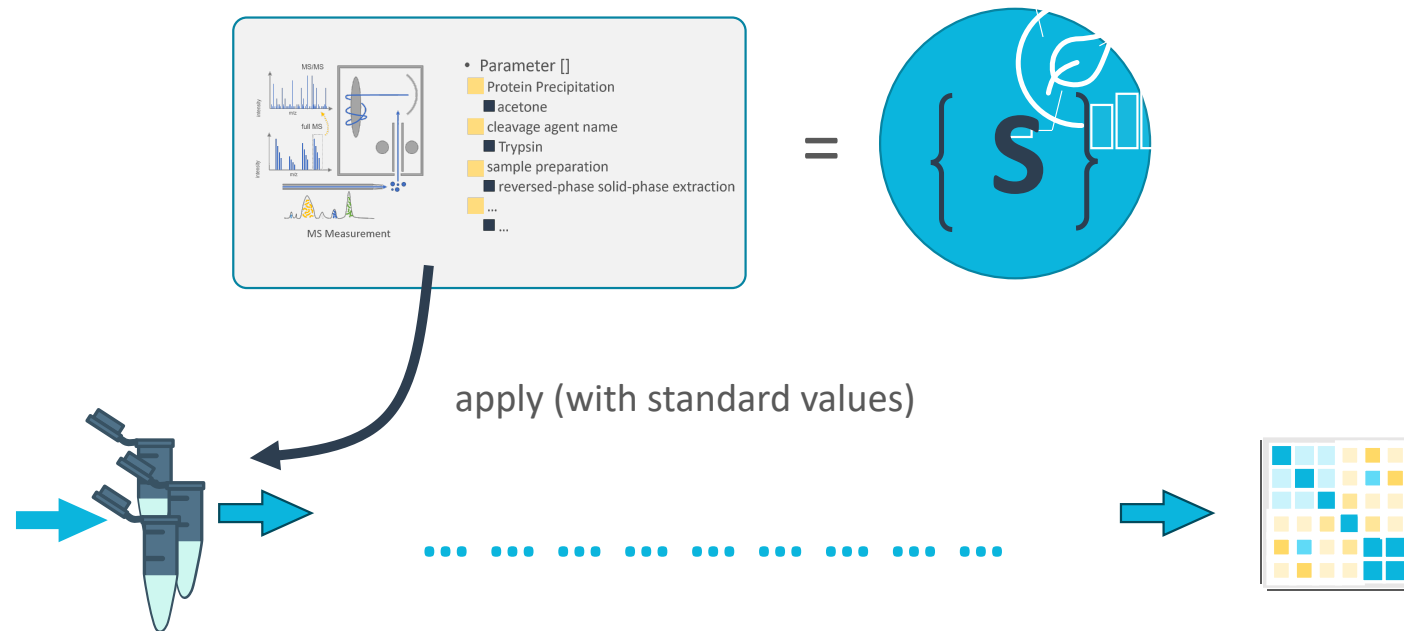
-  Protein Precipitation
 -  acetone
-  cleavage agent name
 -  Trypsin
-  sample preparation
 -  reversed-phase
solid-phase
extraction
- ...

Component []

-  chromatography instrument model
 -  nanoElute2
-  chromatography column model
 -  PepSep C18
1.9u, 25cm x 75um
...



Applying standard procedures to sample record



Hands-on part 2: ARCitect (and Swate)

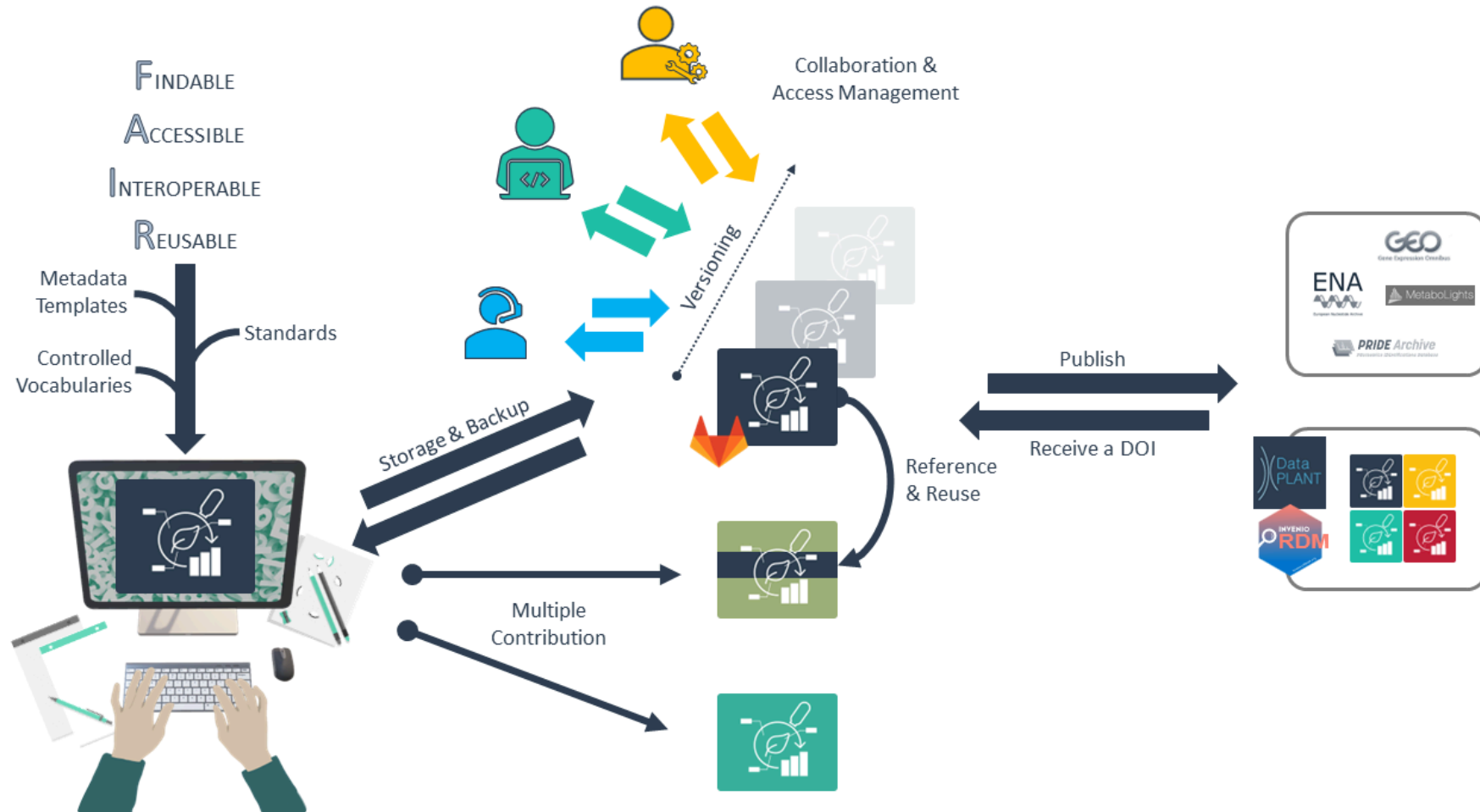
Continue the [Start Here guide](#) in the DataPLANT knowledge base.



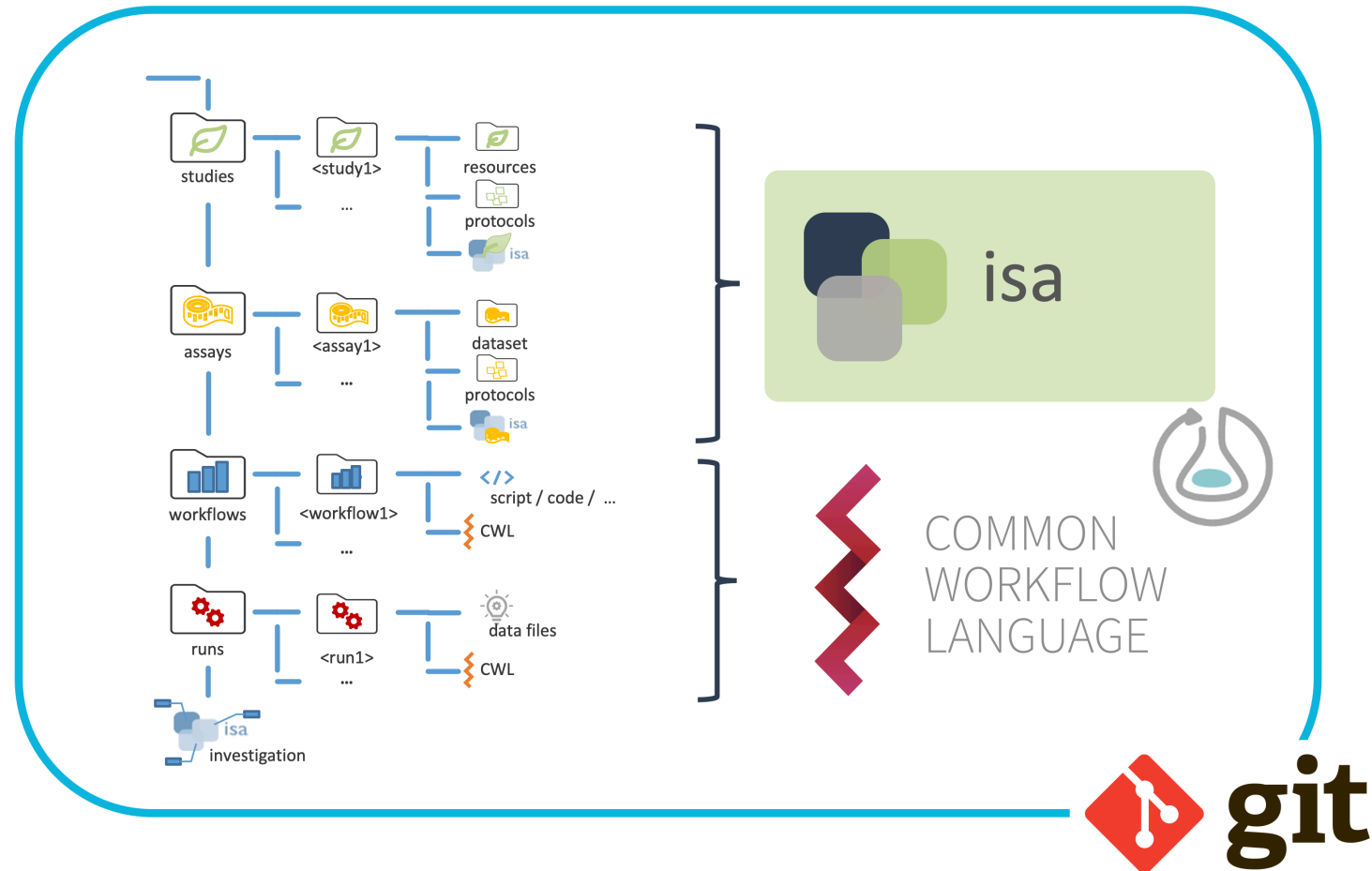
Stop after step [Assay](#)

DataHUB

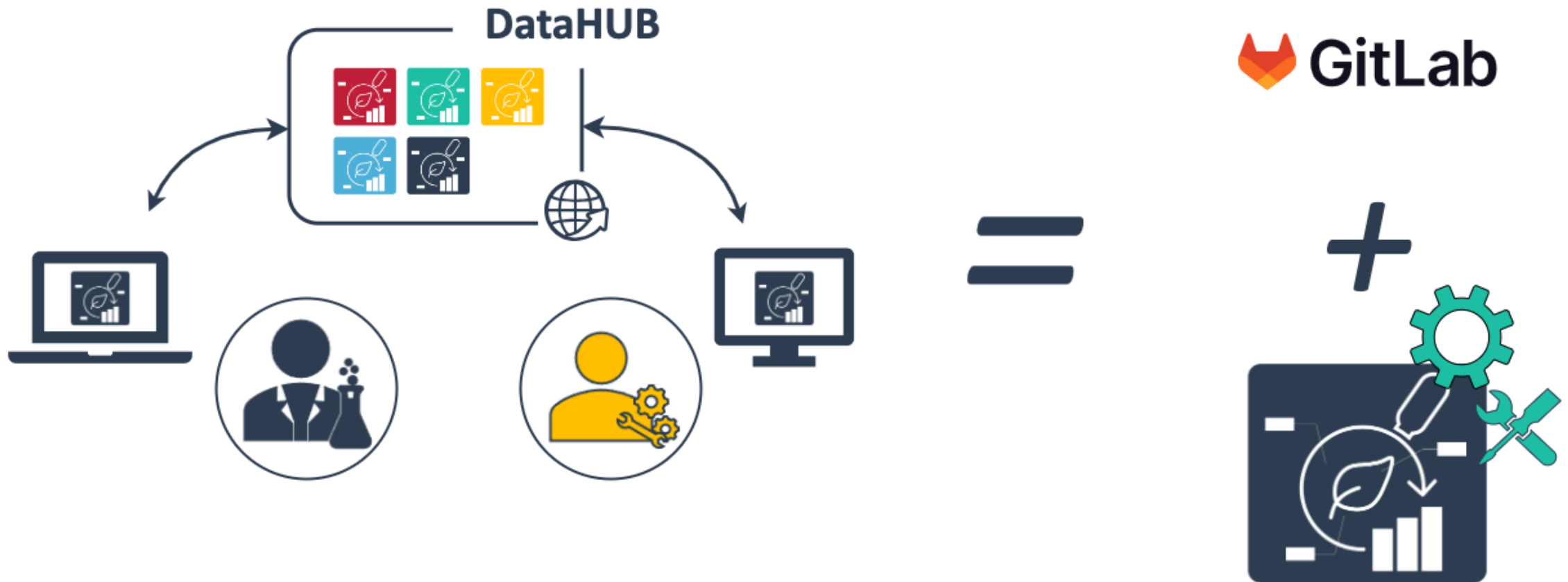
DataHUB overview



ARC builds on standards + Git

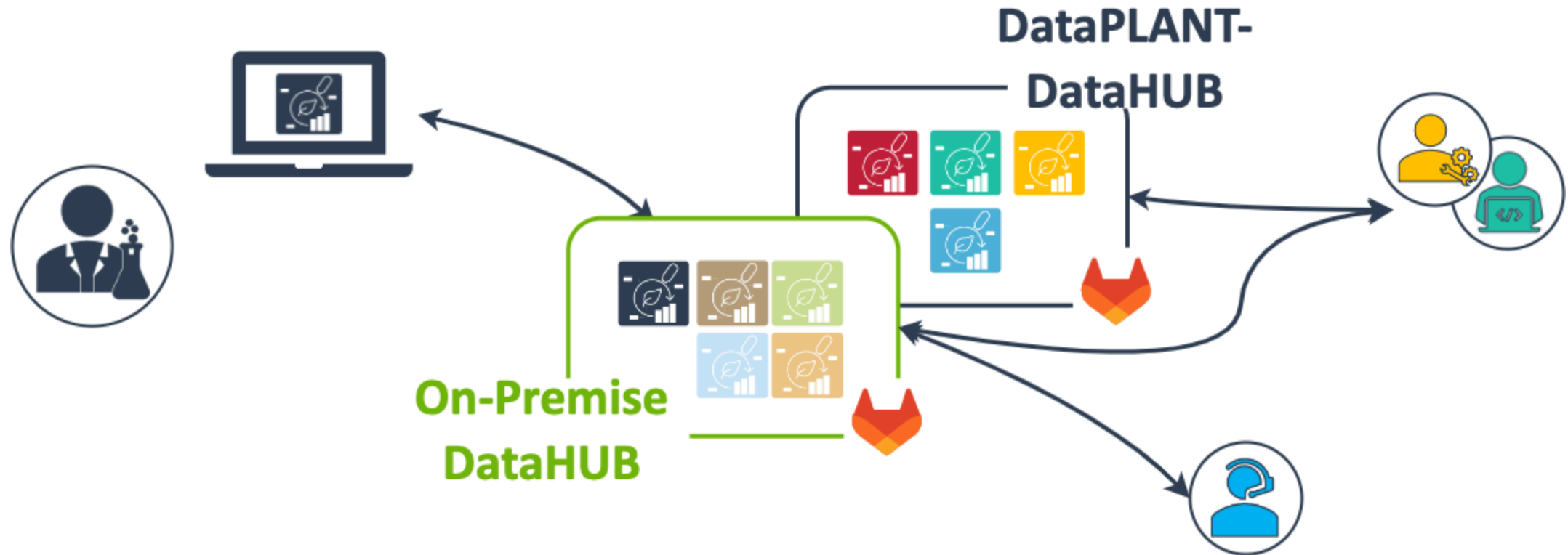


The DataPLANT DataHUB – a GitLab *Plus*



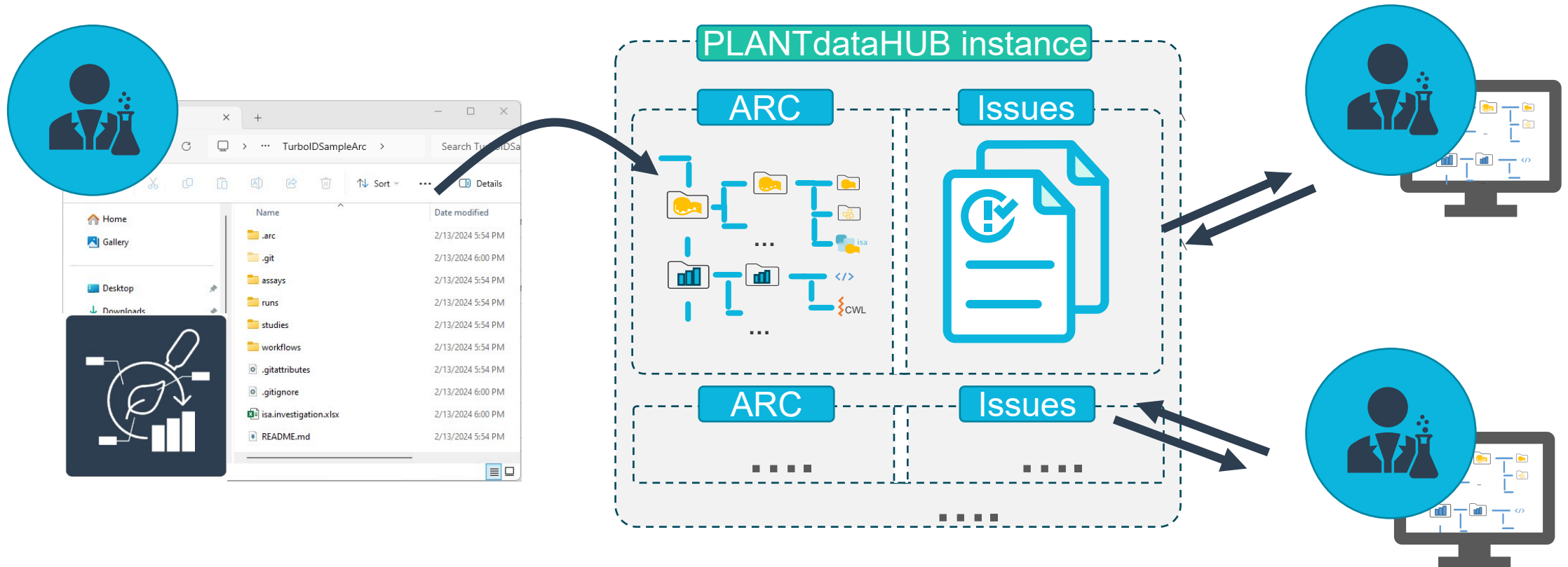
 **GitLab**

On-premise DataHUBs

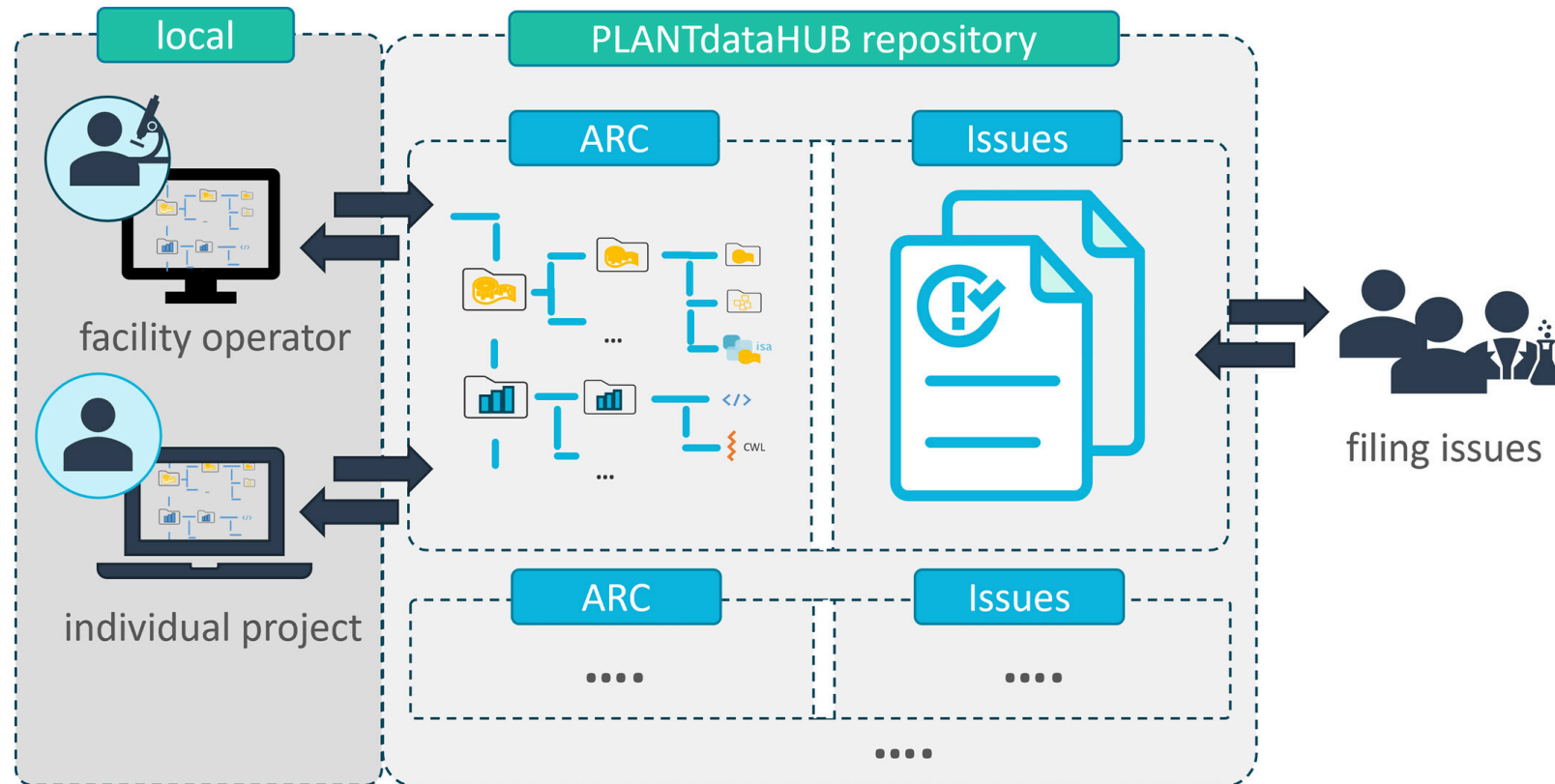


ARC services are available as on-premise option

Using the DataHUB to collaborate

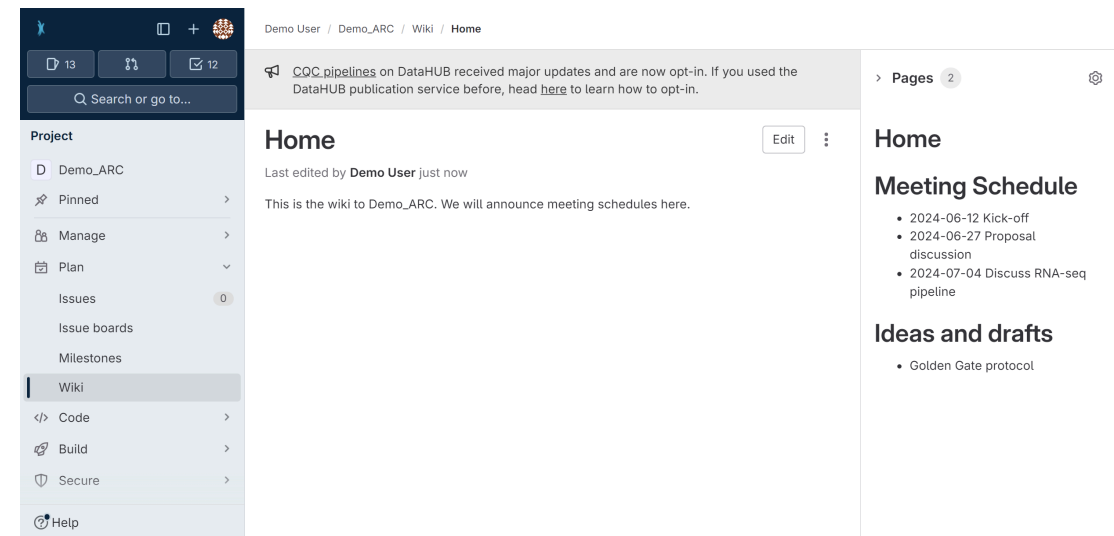


Project management



ARCs come with their own wiki space

- directly associated to your ARC
- same access rights as your ARC
- share meeting minutes or ideas with collaboration partners
- keep ARC clean of files that are not considered "research data"



Namespaces

- Every user has a personal namespace, where they can upload or create new ARCs
- Every group and subgroup has its own namespace

Type	URL	Namespace	Name
A personal ARC	https://git.nfdi4plants.org/brilator/Facultative-CAM-in-Talinum	brilator	Dominik Brilhaus
An group -shared ARC	https://git.nfdi4plants.org/hhu-plant-biochemistry/Samuilov-2018-BOU-PSP	hhu-plant-biochemistry	HHU Plant Biochemistry

💡 **Personal** is not the same as **private**

Visibility

The visibility of ARCs and groups can be managed individually for each ARC or group

Visibility

The visibility of each ARC can be managed in the settings of the ARC



Private – ARC access must be granted explicitly to each user or group.



Internal – ARC can be accessed by any logged in user.



Public – ARC can be accessed without authentication.



By default every ARC and every group is set to **private**.

Permissions & Roles

When inviting new members to an ARC or group, you can choose between different levels.

Permissions & Roles

Roles are assigned when adding a user to an ARC or to a group. This is a very simplified summary.

- Guest** – Can only see the ARC's wiki
- Reporter** – Can read, but not add or edit data
- Developer** – Reporter permissions + can read, add, and edit data
- Maintainer** – Developer permissions + can add new members
- Owner** – Maintainer + can delete ARC, manage memberships and permissions

💡 By default you are **Owner** of an ARC you create or upload to the DataHUB.

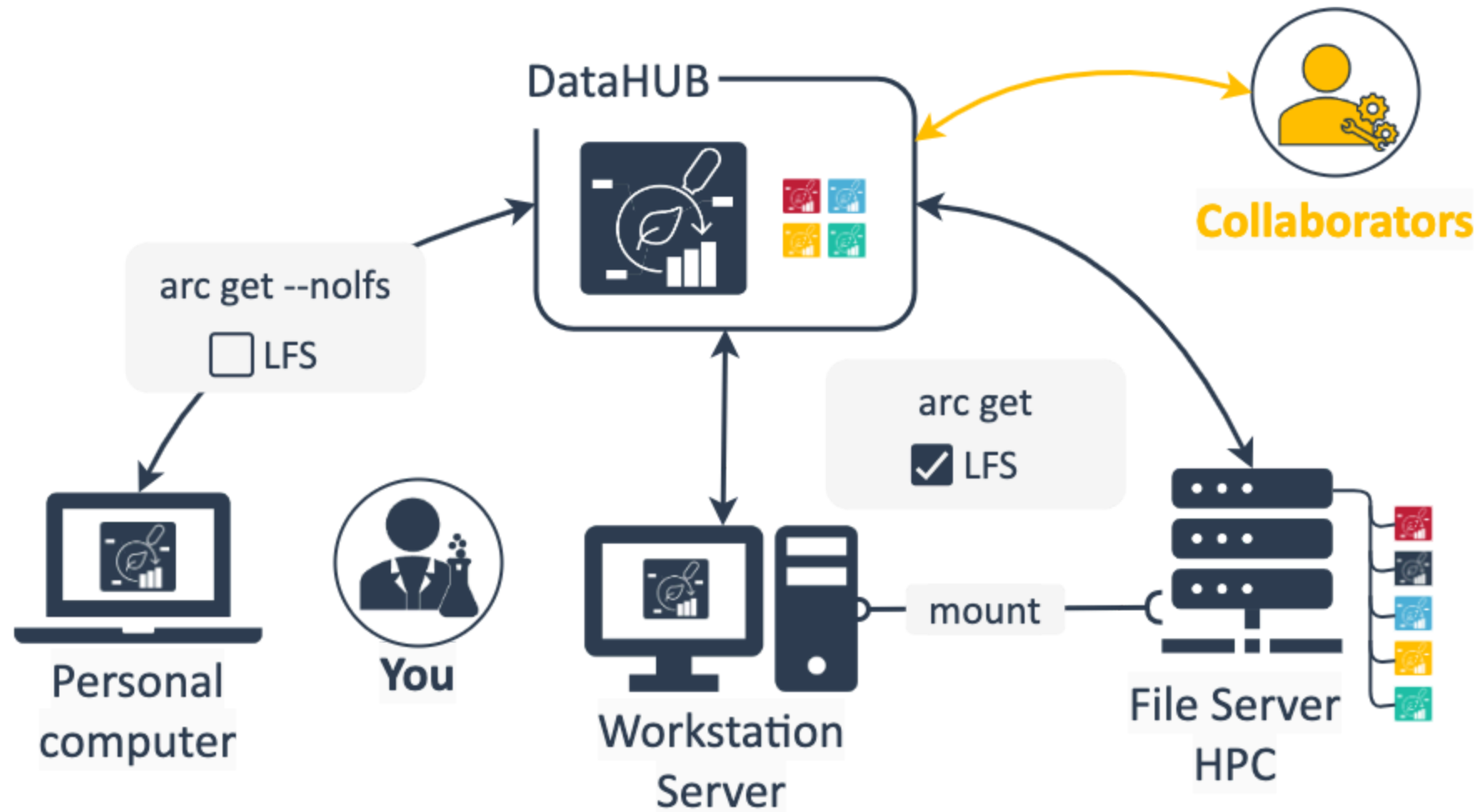
Hands-on part 3: DataHUB

Continue the [Start Here guide](#) in the DataPLANT knowledge base.

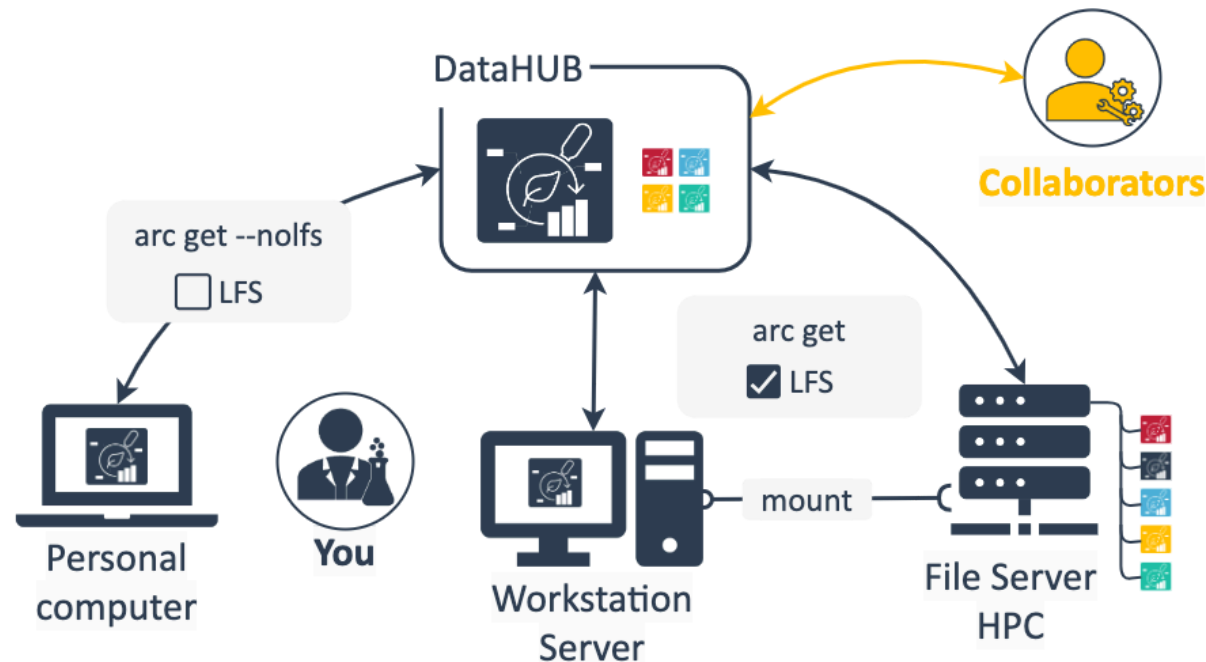
 Until the **Check-point2** step

DataHUB concepts and terminology

Where do I store my ARC?



ARC storage and sharing



- DataHUB as "ground truth" / original clone
- You can sync and communicate all changes to your ARC via the DataHUB

💡 ARCitect and ARC commander provide options to avoid syncing large files (LFS = Large file storage)

Example setup to store and use ARCs

Personal computer

- work on small files
- annotate metadata
- add scripts, protocols

Workstation / Server

- work on large files
- run computations

FileShare

- mount to local machine, sync ARC from there

HPC

- direct connection HPC to DataHUB (depends on security settings)
- or mount to local machine and sync, ARC from there

Projects and Groups are not the same

- "Project" = ARC
- "Groups" = Group of users

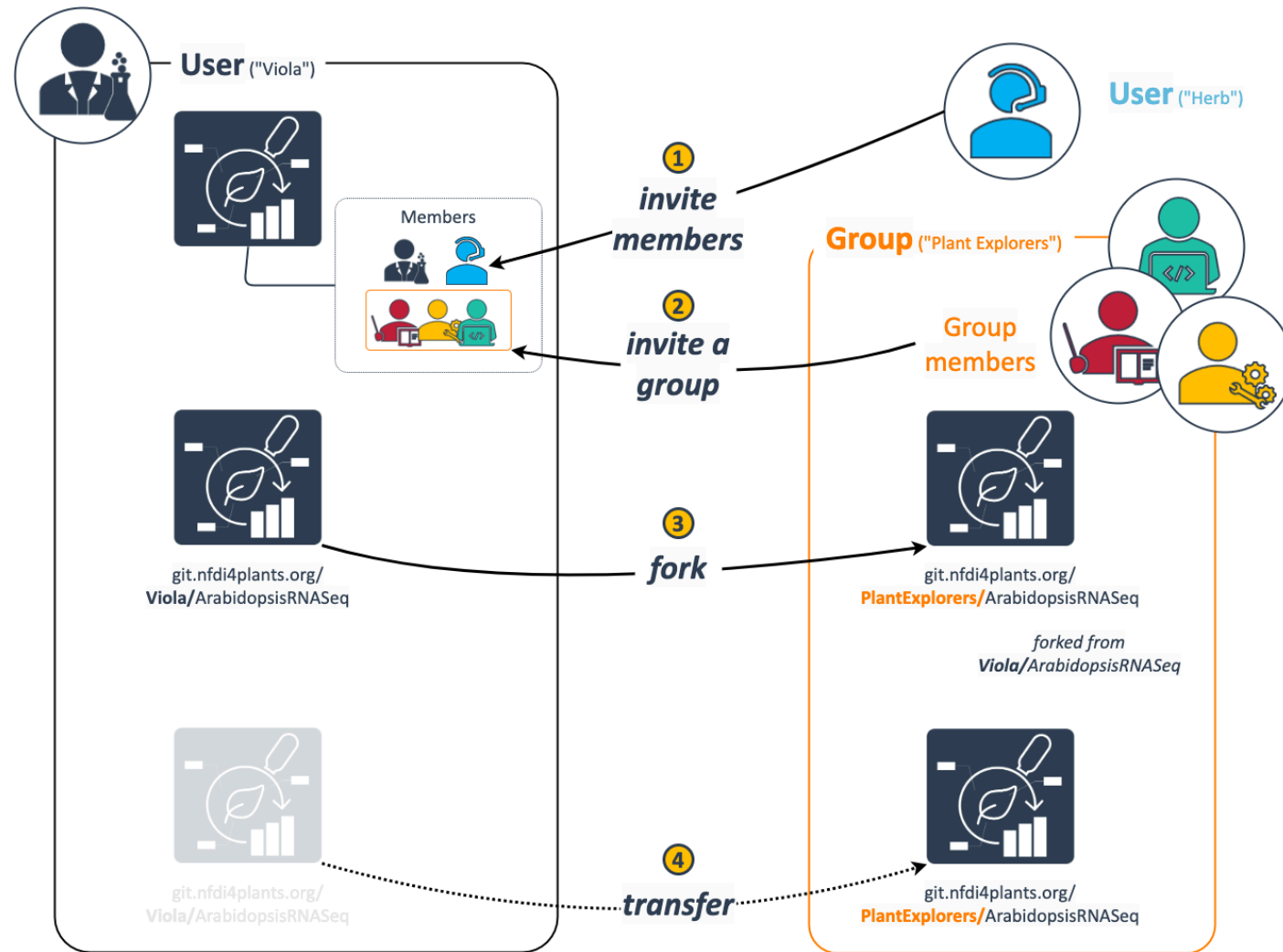
Project = ARC

- In the DataHUB, ARCs are called "projects"; they are the same.
- An ARC can be shared with individual users (invited as "members") or a group.

DataHUB Groups

- A "Group" is a group of users with specific [permissions](#)
- A group can share ARCs
- A group can be invited to an ARC
- Groups can have subgroups

Options to share an ARC via the DataHUB



ARC DataHUB members // ARC Investigation contacts

Project members

You can invite a new member to **Samuilov-2018-BOU-PSP** or invite another group.

Members 18 Groups 1

Account	Source	Max role	Expiration	Activity
Adriano Nunes-Nesi @nunesnesi	HHU Plant Biochemistry by Sebastian Triesch	Maintainer	Expiration date	User created: Jul 05, 2023 Access granted: Jul 10, 2023 Last activity: Aug 09, 2023
Andreas Weber @andreas.weber	HHU Plant Biochemistry by Sebastian Triesch	Maintainer	Expiration date	User created: Mar 10, 2023 Access granted: Jul 31, 2023 Last activity: Sep 11, 2023
Dominik Brilhaus It's you @brilhaus	Direct member by Dominik Brilhaus	Owner	Expiration date	User created: Feb 21, 2022 Access granted: Dec 06, 2023 Last activity: Mar 26, 2024
Franziska Fichtner @franziska.fichtner	HHU Plant Biochemistry by Sebastian Triesch	Maintainer	Expiration date	User created: Aug 11, 2023 Access granted: Aug 11, 2023 Last activity: Aug 11, 2023

DataHUB: ARC members

https://git.nfdi4plants.org/hhu-plant-biochemistry/Samuilov-2018-BOU-PSP/-/project_members

ARCitect

Users / dominikbrilhaus / datahub-dataplant / Samuilov-2018-BOU-PSP

Samuilov-2018-BOU-PSP

- publications
- assays
- runs
- studies
- workflows
- glitignore 1.34 KB
- LICENSE 18.21 KB
- README.md 4.32 KB

Identifier
Samuilov-2018-BOU-PSP

Title

Description

Contacts

Sladjana Samuilov <orcid>	4/10	▼
Nadine Rademacher <orcid>	3/10	▼
Samantha Flachbart <orcid>	3/10	▼
Leila Arab <orcid>	3/10	▼
Saleh Alfarraj <orcid>	3/10	▼
Franziska Kuhnert <orcid>	3/10	▼
Stanislav Kopriva <orcid>	3/10	▼
Andreas P. M. Weber <orcid>	4/10	▼
Tabea Mettler-Altmann <orcid>	3/10	▼

ARCitect: Investigation Contacts

💡 Investigation contacts are not automatically invited as members to the ARC.

elabFTW Introduction

Login into eLabFTW HHU

 eLabFTW

Über Ihre Institution anmelden

Wählen Sie einen Identitätsanbieter aus

HHU-IDM ▾

☒ Anmeldung merken

Anmelden

    | [DATENSCHUTZ-BESTIMMUNGEN](#)

Bereitgestellt von eLabFTW
Seite generiert in 0.00437 Sekunden

eLab-FTW - Dashboard

Dashboard

Welcome Sabrina

Experiments

Create

Resources

Create

Scheduled bookings

Scheduler

Search

Search

Browse by status

RUNNING

SUCCESS

NEED TO BE REDONE

FAIL

Last modified experiments

EXP02 | INV024 | SDY001 | Host Depletion 20 hours ago

EXP01 | INV024 | SDY001 | Seed Sterilization 20 hours ago

INV020 | SDY003 | EXP009 5 days ago

INV020 | SDY003 | EXP008 5 days ago

INV020 | SDY003 | EXP007 6 days ago

INV020 | SDY003 | EXP006 12 days ago

Browse by category

SOFTWARE

EQUIPMENT

PRIMER LIST

LICHEN SHEET

PLASMID COLLECTION

LICHEN COLLECTION

PLANT COLLECTION

SOP

FORMBLATT Z

EQUIPMENT-VARIOUS

MEDIA AND SOLUTIONS

PLANT SHEET

PLASMID SHEET

MANUAL

METHOD

LIST OF INVESTIGATION

STUDY

INVESTIGATION

Last modified resources

STUDY INV020 | SDY003 | Trichome analysis (Bachelor thesis) 24 days ago

INVESTIGATION INV020 | Project Morelloid Clade 27 days ago

LIST OF INVESTIGATION List of INVESTIGATIONs 27 days ago

INVESTIGATION INV024_CL 1 month ago

METHOD MTH0021 | ONT SQK-16S114.24 for Seed Microbiota 2 months ago

METHOD MTH0023 | SQK-NBD114.96 + Adaptive Sampling for Microorganisms 2 months ago

Powered by eLabFTW 5.5.8

Made with by Deltabiot

eLabFTW - Experiment panel

The screenshot shows the eLabFTW Experiments panel. The header includes the eLabFTW logo and navigation links: EXPERIMENTS, RESOURCES, TEAM, SEARCH, and DOCUMENTATION. A search bar, notification bell, help icon, and user profile icon are on the right. The main section is titled 'Experiments' with a 'Create' button. Below the title is a filter bar with 'Expand all - Select all', a table view icon, and buttons for 'Scope', 'Filters', and 'Sort'. The table has columns: Date, Title, Next step, Category, Status, Tags, Rating, and Owner. Two experiments are listed:

Date	Title	Next step	Category	Status	Tags	Rating	Owner
☐ 2025-01-09	INV0001_A1_Sugar measurement			SUCCESS	A.thaliana INV001 sugar		Sabrina Zander
☐ 2025-01-09	INV0001_A1_Sugar Extraction			RUNNING	A.thaliana INV001 sugar		Sabrina Zander

A 'Load more' button is at the bottom of the table. The footer contains social media icons, links to PRIVACY POLICY and TERMS OF SERVICE, and a note: 'Powered by eLabFTW 5.1.12 Made with ♥ by Deltablot'.

Default "working area" for protocols of daily lab work

eLabFTW - Resource panel

eLabFTW

EXPERIMENTS

RESOURCES

TEAM

SEARCH

DOCUMENTATION

Resources

Create

Expand all - Select all

Scope

Filters

Sort

Date	Title	Next step	Category	Status	Tags	Rating	Owner
☐ 2025-01-09	INV0001_S1_AthalianaColdStress		STUDY		A.thaliana INV001		Sabrina Zander
☐ 2025-01-09	Growth Protocol for Arabidopsis thaliana		METHODE		A.thaliana		Sabrina Zander
☐ 2025-01-09	INV0001_A.thaliana_ColdStress_Sugar		INVESTIGATION		A.thaliana INV001		Sabrina Zander

Load more

X

PRIVACY POLICY

TERMS OF SERVICE

Powered by eLabFTW 5.1.12
Made with ♥ by Deltablot

Place to categorize content

eLabFTW - Experiment information

INV0001_A1_Sugar Extraction

signature
timestamp
different export functions

lock request action

pin

More options

status

tags

sharing

Transfer ownership
See revisions
See changelog
Archive/Unarchive
Delete

Started on 09.01.2025

ID 185 Custom ID Get next

Category Assay

Status Running

Tags A.thaliana INV001 sugar Add a tag

Suggested tags

Visibility Only members of the team

Can write Only owner and admins

MAIN TEXT Last saved: 6 minutes ago

eLabFTW - Tag usage

Very individual, some examples:

- Investigation ID
- Subgroup
- Organism /part (e.g. A.thaliana, leaf, root, mitochondria)
- Goal (strain generation, plasmid generation)
- Methode (e.g. transformation, western blot, extraction, biolector)
- Genes/ Proteins (e.g. mKate2, eGfp, Rrm4)

Use keywords you would search/filter for



Make rules for tag usage

eLabFTW - Extra fields

 Add an extra field ×

Group 

Undefined group 

Field type

Text 

Text

Dropdown menu

Radio buttons

Date

Date and time

Email

Time

Checkbox

Number

Experiment

Resource

User

URL

Close

Save

eLabFTW - Extra fields examples

Dropdown	URL	Number	Date
Field type Dropdown menu Name (required) * Antibiotic used Description Select the antibiotic used Choices Kanamycin Streptomycin Ampicillin Add another choice ☐ Allow selection of multiple values?	EXTRA FIELDS UNDEFINED GROUP OMERO dataset https://omero-cai.hhu.de/webclient/?show=dataset-12881	Extra fields End date 06 / 09 / 2021 Magnification 20X Pressure (Pa) 12 Wavelength (nm) 488 405 647	Field type Number Name (required) * Drug concentration Description Indicate the drug concentration Available units ng/mL µg/mL mg/mL Add another choice

eLabFTW - Experiment categories examples

- Study
- Assay
- Discussion
- Teaching
- Demo
- R&D
- Production

eLabFTW - Categories in Resources

Use Resources to categorize content & manage lab.

Use unique identifier & naming conventions (e.g. Methode MTH0001, Media MED0001, Plasmid PL0001).

- List of Investigation
- Investigation
- Plant sheet
- Plasmid sheet
- Equipment
- Collections
- Methods
- Manuals
- Media & Solution
- Formblatt Z

eLabFTW - Scheduler

SCHEDULER MEMBERS EMAIL PROCUREMENT REQUESTS BETA

Filter by category Equipment - Zeiss AXIO Observer Z1 Inverted Fluorescence Microscope x

X EQUIPMENT Zeiss AXIO Observer Z1 Inverted Fluorescence Microscope

< > today 6 - 12 Jan 2025 week list month

	Mon 06/01	Tue 07/01	Wed 08/01	Thu 09/01	Fri 10/01	Sat 11/01	Sun 12/01
00							
01							
02							
03							
04							
05							
06					6:00 - 12:30 (Sabrina Zander)		
07							
08							
09							
10							
11							
12							

eLabFTW - Booking parameter

You can modify booking parameter for reach Resource

Zeiss AXIO Observer Z1 Inverted Fluorescence Microscope

Create

←

Started on

09.01.2025

ID

115

Custom ID

Get next

Category

Equipment

Status

Not set

Tags

Add a tag

► Suggested tags

Modify booking parameters

Modify procurement parameters

Transfer ownership

See revisions

See changelog

Archive/Unarchive

Delete

eLabFTW - Templates

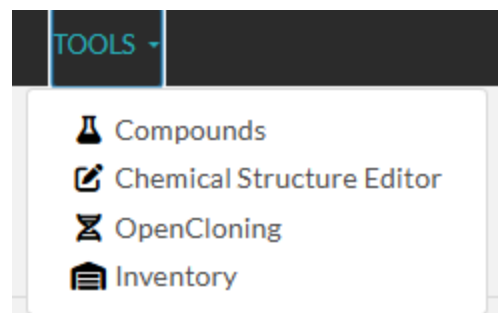
Create templates for experiments you often use

- „Skeleton“ of a real experiment
- Select „Templates“ from User menu
- You can have different permissions for the template itself and for Experiment that will be created from that template
- You can add tags, steps, links to it (e.g. link to project)
- Can also be imported directly in your text (allows merging several templates in one experiment)

eLabFTW - Tools

Different tools are implemented in eLabFTW

- Compounds: Chemical entries associated with their properties
- Chemical structure editor: Draw chemical compounds
- Open Cloning: can be used for plasmid cloning
- Inventory: Add inventory and locations



eLabFTW - Tipps and tricks

- Use # to link experiments/resources in the text
- Use .gb files for plasmid cards (can be opened in a plasmid viewer in elabFTW)
- Use versions, possibility to upload new versions of files/methods

eLabFTW - Support

General

<https://github.com/elabftw/elabftw>

<https://www.elabftw.net/> (Newsletter)

HHU specific

elabftw@hhu.de

https://rocketchat.hhu.de/channel/elb_eln

<https://wiki.hhu.de/display/ZIM/eLabFTW>

elab2ARC Documentation

<https://nfdi4plants.github.io/nfdi4plants.knowledgebase/resources/elab2arc/>

Working Scenario

1. Use ARCs to

- Organize and store (raw) data locally
- Share and communicate data with collaboration partners
- Annotate and prepare data for publication

2. Use elabFTW to

- Document daily lab work „on the fly“
- Organize lab routines with colleagues

eLabFTW & ARC Connection

Aim

- Reducing duplication/ additional workload
- Transformation of data from elabFTW into ARC
- User-friendly, easy implementation

Challenges

- ELNs are unstructured; automated translation and interoperation into well-structured ARC challenging

Solution

- Start with automated transition of data without
 - metadata mapping,
 - structured metadata extraction (now optional with LLM),
 - re-structuring of elabFTW content

Conversion

Experiment1 DNA Extraction

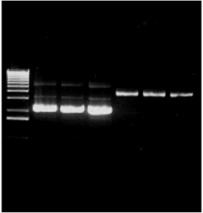
▼ MAIN TEXT

Experiment Protocol:

Goal: Extract DNA

Procedure: Use Kit

Result: Gel picture



ATTACHED FILE



[Abbildung-14-Agarose-Gel-Elektrophorese-Unterschiedliche-Konzentration-an-plasmidischer.png](#) 12.94 KiB · 2024-09-10 11:29:25

Click to add a comment

elab2ARC



▼ assays

▼ **Experiment1_DNA_Extraction**

▼ dataset

 0_Abbildung-14-Agarose-Gel

 readme.md

▼ protocols

 eLabFTW_protocol.md

isa_assay

Conversion

The elab2ARC tool will automatically convert your elabFTW experiments into ARC format

- create a new assay folder with eLabFTW experiment name as assay name
- create the assay folder structure (dataset/protocols/isa.assay)
- convert experiment main text into a .md file and store it in the protocol folder
- add all attachments of the eLabFTW experiment into the dataset folder
- enter name/email/affiliation of the eLabFTW experiment metadata into the isa.assay sheet

optional LLM usage

LLM modals extract metadata from free-text into machine-readable ISA tables

Assay Metadata

Identifier

Experiment_1_Bacterial_Cultivation

Title

eLabFTW_protocol_40_Experiment_1_Bacterial_Cultivaelab2arc

Description

Bacterial cultivation of _E. coli_ Strain K-12 (n=10)

Measurement Type

Search terms...

Technology Type

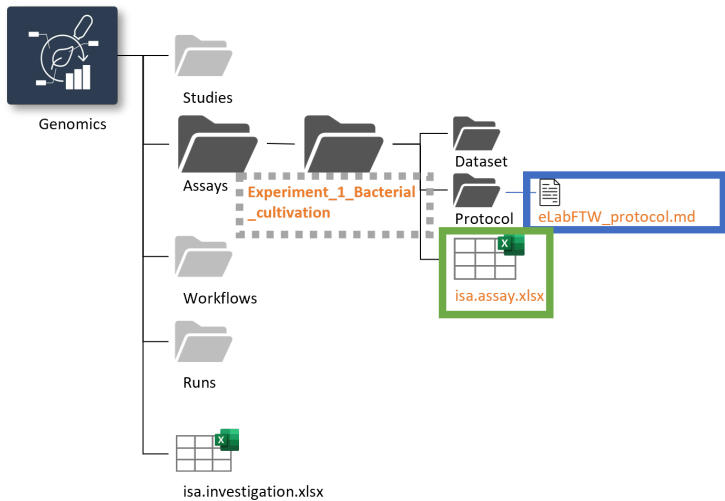
Search terms...

Technology Platform

Search terms...

Assay

LLM disabled



	Input [Source Name]	Characteristic [Organism]	Characteristic [strain]	Characteristic [Initial OD600]
1	Sample_1	E. coli	K-12	0.051 OD600
2	Sample_2	E. coli	K-12	0.049 OD600
3	Sample_3	E. coli	K-12	0.053 OD600
4	Sample_4	E. coli	K-12	0.050 OD600
5	Sample_5	E. coli	K-12	0.052 OD600
6	Sample_6	E. coli	K-12	0.051 OD600
7	Sample_7	E. coli	K-12	0.048 OD600
8	Sample_8	E. coli	K-12	0.050 OD600
9	Sample_9	E. coli	K-12	0.053 OD600
10	Sample_10	E. coli	K-12	0.049 OD600

	Input [Source Name]	Protocol REF	Parameter [Temperature]	Parameter [Shaking speed]	Parameter [Incubation time]	Parameter [Medium]
1	Sample_1	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
2	Sample_2	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
3	Sample_3	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
4	Sample_4	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
5	Sample_5	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
6	Sample_6	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
7	Sample_7	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
8	Sample_8	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
9	Sample_9	protocols/eLa...	37 °C	180 rpm	16 h	LB broth
10	Sample_10	protocols/eLa...	37 °C	180 rpm	16 h	LB broth

elab2ARC tool Hands-on

tutorial: <https://nfdi4plants.github.io/nfdi4plants.knowledgebase/workshops/2026-fdm-werkstatt/2026-fdm-werkstatt-elab2arc/>

- usage with demo data

elab2ARC Tool

Tool: <https://nfdi4plants.github.io/elab2arc/>

Documentation:

<https://nfdi4plants.github.io/nfdi4plants.knowledgebase/resources/elab2arc/>

Issues/Problems:

<https://github.com/nfdi4plants/elab2arc>

Contributors

If not referenced otherwise, figures and slides presented here were created by members of DataPLANT (<https://nfdi4plants.org>).

Additional slides were contributed by

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